



A shock tube and chemical kinetic modeling study of the pyrolysis and oxidation of butanols

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ABSTRACT

The pyrolysis and oxidation of all four butanols (*n*-, *sec*-, *iso*- and *tert*-) have been studied at pressures from 1 to 4 atm and temperatures of 1000–1800 K behind reflected shock waves. Gas chromatographic sampling at different reaction times varying from 1.5 to 3.1 ms was used to measure reactant, intermediate and product species profiles in a single-pulse shock tube. In addition, ignition delays were determined at an average reflected shock pressure of 3.5 atm at temperatures from 1250 to 1800 K. A detailed chemical kinetic model consisting of 1892 reactions involving 284 species was constructed and tested against species profiles and ignition delays. The little-known chemistry of enols is included in *this work* to explain the temperature dependence of acetaldehyde produced in the thermal decomposition of isobutanol.

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1. Introduction

Among the greatest problems facing humankind today is climate change due to global warming, which has largely been brought about by the combustion of fossil fuels [1]. In addition, diminishing energy resources and the security of energy supply in the context of international political instability are becoming increasing causes for concern. It is clear however, that the combustion of fossil fuels is still the dominant means of energy production globally, and this dependence will not be easily severed.

Though the overall benefit of a biofuel economy remains a contentious question, biofuels ostensibly offer numerous advantages – primarily that the net carbon dioxide emissions associated with their combustion are lower than in the case of fossil fuels. They are also a renewable energy resource, as well as implying a diversification of energy supply. However, the chemistry of their combustion must necessarily differ from that of conventional fossil fuels, due principally to the fact that biofuels are usually oxygenated compounds rather than hydrocarbons [2].

Although the combustion of ethanol, which is already in common use as a transport fuel, has been well investigated, many new biofuels are emerging and the consequences and optimum conditions for burning these types of fuel are therefore unknown. This is where detailed studies, such as *this work*, of the combustion chemistry of these compounds comes into play.

The most widely used method for the production of bio-alcohols is fermentation – the conversion of sugars into alcohol by yeast or other micro-organisms. The ABE process is used to produce a mixture of acetone, butanol and ethanol via the bacterial fermentation of starch, usually using the organism *Clostridium acetobutylicum*, although the associated drawback of this method is that the products must then be separated from one another. Recent developments in butanol production have included strategies for improving the yield and method of isolation of the products in the ABE process, as well as reducing the toxicity of the alcohol to the bacterial culture [3].

These ‘first generation’ production methods suffer from a number of flaws – chief amongst them is the fact that the organisms involved cannot tolerate alcohol concentrations greater than about 15% and cease to function above this level. Also, the isolation of alcohol from the watery ‘soup’ produced is quite difficult and requires distillation, followed by chemical treatment to remove the final amounts of water. This second step is necessary in the case

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of ethanol, for example, because it is infinitely miscible with water, which itself does not mix with petrol, and so any remaining water will prevent the alcohol and petrol from blending properly.

Atsumi et al. [4] have described a non-fermentative pathway ('metabolic engineering') for synthesizing higher alcohols, including isobutanol, *n*-butanol, 2-methyl-1-butanol, 3-methyl-1-butanol and 2-phenyl ethanol – in comparison to ethanol, the higher alcohols have a greater energy density and are not as hygroscopic, which makes them suitable candidates to potentially replace it.

Much practical work has been carried out particularly on *n*-butanol in internal combustion engines [5–7]. For example, in a direct comparison of 10% blends in gasoline Wallner et al. [8] have found that butanol performs equally as well as ethanol in a direct-injection spark-ignition engine from the standpoint of regulated emissions and combustion, but with a decrease in fuel consumption.

The development of bio-butanol as a competitive biofuel is therefore well underway, but it yet remains to examine the chemistry of its combustion in order to optimize its performance in internal combustion engines and assess the possibility of consequent harmful emissions. Recent work [9–13], including that reported here, has set about this task.

Studies of all four C_4H_9OH alcohols that is, normal (*n*-), secondary (*sec*-) and tertiary (*tert*-) butanol in addition to 2-methyl propanol (isobutanol), shown in Fig. 1 are far fewer by comparison. McEnally and Pfefferle [14] compared the effect of doping a methane/air flame with small amounts of each butanol in turn and measured the quantities of C1–C12 hydrocarbons formed and major stable species.

Yang et al. [15] investigated low-pressure premixed laminar butanol flames using tunable synchrotron vacuum-ultraviolet radiation photoionization and molecular beam mass spectrometry to identify both transient and stable species.

Moss and co-workers [16] have measured ignition times for all four butanols behind reflected shock waves and carried out chemical kinetic modeling showing that hydrogen atom abstraction from the most reactive fuels, *n*-butanol and isobutanol, is principally responsible for consumption of the alcohol.

Grana et al. [17] developed a reaction mechanism for all four butanols, and carried out a validation of their kinetic mechanism by comparing the model prediction with a wide range of experimental data such as (i) time-dependent water concentration profiles from *tert*-butanol pyrolysis in a shock tube [18], (ii) time-dependent concentration profiles of *tert*-butanol and products from *tert*-butanol oxidation in a flow reactor [19], (iii) concentration profiles of *n*-butanol and products from *n*-butanol oxidation in a jet stirred reactor [11], (iv) ignition delay times in shock tubes [10,16], and (v) the structures of *n*-butanol [11,17] and isobutanol [17] counterflow non-premixed flames.

Van Geem et al. [20] generated a reaction mechanism of *n*-, *sec*- and *tert*-butanols, and validated it against (i) ignition delay times [16], (ii) doped methane flame concentration profiles [14] and

(iii) their own pyrolysis data. More recently they have used a reaction mechanism generator to construct a detailed mechanism for *n*-butanol [21].

In this study we have carried out species profile and ignition delay measurements of all the butanols heated behind reflected shock waves at pressures from 1–4 atm, in the temperature range 1000–1800 K using a magic-hole, single-pulse shock tube and a standard shock tube and assembled a detailed chemical model to account for these and other recent experiments.

2. Experimental

Samples of *n*-, *tert*-, *sec*- and isobutanol fuels, 98.0%, 99.0%, 99.0%, and 99.0% pure were supplied by Wako Pure Chemical, and de-gassed through a series of freeze–pump–thaw cycles, after which no more gas was observed to escape as the solid thawed. The argon, 99.9999% pure was supplied by Iwatani. Mixtures were prepared using the method of partial pressures and their compositions are provided in Table 1. The incident shock velocity at the end wall was used to calculate the temperature and pressure of the mixture behind the reflected shock wave using the equilibrium program Gaseq [22].

Two shock tubes were used in this study, the first described herein is based at the National Defense Academy in Japan, while the second resides at NUI Galway. The first shock tube (Single-Pulse Shock Tube: SPST) has an internal diameter of 4.1 cm and has been described in detail previously by Hidaka et al. [23,25–29] In single-pulse mode the reacted gas mixtures are rapidly quenched by the rarefaction wave and then extracted into a pre-evacuated vessel (50 cm³) through a valve near the endplate. The reacted gas mixtures were analyzed by three serially connected gas chromatographs each having a thermal conductivity detector (TCD) [26,27].

The gas chromatographic analyses were carried out as follows: a Shimadzu GC-8A with a 2 m Sebacitrile column was used at 75 °C to determine concentrations of *n*-, *tert*-, *sec*-, and *iso*- C_4H_9OH , CH_3COCH_3 , C_2H_5CHO (propanal) and CH_3CHO (acetaldehyde). The retention times for *n*, *iso*-butylaldehyde and methyl ethyl ketone as possible products of pyrolysis were checked, and any peak for these species was not measured under our experimental conditions. A Shimadzu GC-8A with a 2 m Porapak Q column coupled to a 2 m Unibeads 1S column was used to determine the concentrations of C_2H_6 (ethane), C_2H_4 (ethylene), C_2H_2 (acetylene), C_3H_8 (propane), C_3H_6 (propene), C_3H_4 -a (allene), C_3H_4 -p (propyne), C_4H_8 -1 (1-butene), and iC_4H_8 (isobutene); columns were heated at a rate of 3 °C/min from 50 °C to 130 °C. A Shimadzu GC-3BT with 2 m Molecular Sieve 5A column at

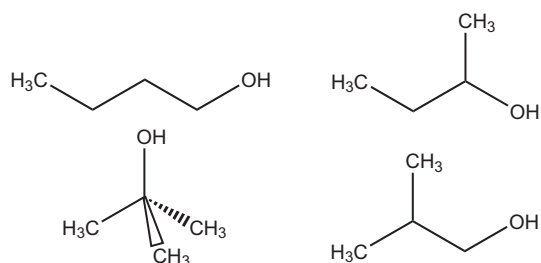


Fig. 1. Structure of butanol isomers.

Table 1
Composition of mixtures (mol %).

Mixture	<i>n</i> - C_4H_9OH	<i>tert</i> - C_4H_9OH	<i>sec</i> - C_4H_9OH	<i>iso</i> - C_4H_9OH	O ₂	Ar
A	1.5					98.5
B		1.0				99.0
C			1.5			98.5
D				1.5		98.5
E		1.0			6.0	93.0
F		1.0			12.0	87.0
G		1.0			24.0	75.0
H			1.0		3.0	96.0
I			1.0		6.0	93.0
J			1.0		12.0	87.0
K				1.0	3.0	96.0
L				1.0	6.0	93.0
M				1.0	12.0	87.0

50 °C was used to determine the concentrations of CH₄ and CO. Helium was used as a carrier gas. The output signal from each gas chromatograph was introduced into Shimadzu Chromatopac C-R8A data processors.

An effective heating or reaction time, t_e , which was defined as the time between the arrival of the reflected shock wave and the point at which the reflected shock pressure had fallen by 20% was determined using a method described previously [25,26]. On the assumption that the process is an adiabatic expansion and that the chemistry is frozen, it can be calculated that the temperature drops by 8.5% from its initial value at the effective heating time.

Given that single-pulse shock tubes have cooling rates [23] of $6.6 \times 10^5 \text{ K s}^{-1}$, it can be assumed that the reaction was frozen at the effective heating time, and hence that the concentrations of those species determined by gas chromatography can be compared with those from a simulation. The validity of effective heating time and cooling rate was tested for N₂O pyrolysis previously [23]. The experimental uncertainties of the concentrations are less than 10% except for *n*-C₄H₉OH due to condensation and strong adsorption to the surface of gas collection line. The estimated experimental uncertainty of the concentration of *n*-C₄H₉OH was less than 50%. The other oxygenates species do not show strong scatter, the case of acetaldehyde is seen in Fig. 1 of [24].

The second shock tube (UV Emission Shock Tube: UEST), a standard-type based at NUI Galway, consists of a test section measuring 6.22 m in length, with an internal diameter of 10.24 cm, and a barrel-shaped driver section measuring 53 cm in length. The two sections are separated by a polycarbonate diaphragm, which bursts when forced into contact with a cross-shaped cutter due to the pressure differential. Four pressure transducers located along the last half meter of the tube were used to determine the velocity of the incident shock wave. The velocity at the end plate was determined through linear extrapolation, so as to take attenuation of the shock wave into account.

The pressure at the end plate was monitored by a pressure transducer (Kistler, model 603B). Light emission at 431 nm, which is emitted from excited CH* radicals, was detected through a fused silica window embedded in the end-plate using a photo-detector (Thorlabs Inc. PDA55-EC) and a narrow band-pass filter centered at 430 nm with a full-width half-maximum of 10 nm. The ignition delay time was defined as the interval between the rise in pressure due to the arrival of the shock wave at the end plate and the maximum rate of rise of the emission signal.

3. Model

The thermodynamic data for the butanol isomers and related radicals were calculated using the THERM program of Ritter and Bozzelli [30], based on the group additivity methods developed by Benson [31], and are provided in Table 2. The thermodynamic data for the butanol isomers were compared with the recommended values in the NIST WebBook [32] with which they are in good agreement. NASA polynomials covering a wider range of conditions for the entropy, specific heat at constant pressure and enthalpies are provided in Supplementary Information.

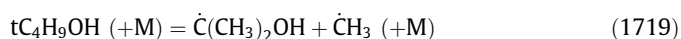
We developed a reaction mechanism for butanol isomers by adding primary reactions of *tert*-, *sec*- and isobutanol and related radicals reactions to the *n*-butanol reaction mechanism proposed by Black et al. [10]. Several reactions concerning *n*-butanol were also updated based on the same rule applied to the other butanol isomers. This update does not have a strong effect on the model prediction of species profile of *n*-butanol pyrolysis behind reflected shock wave. There are few rate constants measured for the primary reactions of butanol isomers, the remainder have been estimated in a number of ways detailed below. A full listing of the reaction mechanism is available online at <http://c3.nuigalway.ie/mechanisms.html>. Particular reactions and rate constants for butanol isomers are provided in Table 3.

3.1. Unimolecular decomposition

3.1.1. *tert*-Butanol

The *tert*-butanol sub-mechanism includes the two possible classes of unimolecular decomposition reaction, including simple bond scission and water elimination.

The high-pressure limit rate constant for simple bond scission, reaction (1719):



was obtained from work by Tsang [36] on 3,3-dimethylbutan-2-ol (3,3-DMB-2-ol) and 2,3-dimethylbutan-2-ol (2,3-DMB-2-ol). These two alcohols have similar C–C bond scission reactions to *tert*-butanol.

Those for the other reactions:

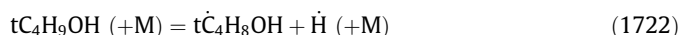
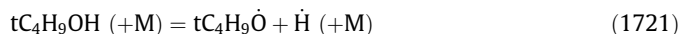
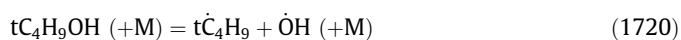


Table 2
Thermochemical data of butanol isomers and related radicals.

	ΔH_f° (298.15 K)	S° (298.15 K)	C_p (cal mol ^{−1} K ^{−1})				
	(kcal mol ^{−1})	(cal mol ^{−1} K)	300 K	500 K	800 K	1000 K	1500 K
sC ₄ H ₉ OH	−69.85	84.62	26.90	39.79	53.15	59.06	67.66
sC ₄ H ₉ OH ^a	−70.50	84.94	27.07	39.59	52.65	58.63	66.04
sC ₄ H ₈ OH-1	−28.45	89.09	26.13	36.97	48.97	54.55	62.68
sC ₄ H ₈ OH-2	−21.25	89.06	25.40	36.69	48.68	53.53	61.23
sC ₄ H ₈ OH-3	−20.85	88.04	26.13	37.88	50.62	56.28	64.07
sC ₄ H ₈ OH-M	−19.45	88.04	26.13	37.88	50.62	56.28	64.07
sC ₄ H ₉ O	−18.12	81.79	25.92	38.18	51.05	56.69	64.48
iC ₄ H ₉ OH	−67.82	83.40	26.42	39.50	52.92	58.93	68.47
iC ₄ H ₉ OH ^a	−67.80	83.65	–	–	–	–	–
iC ₄ H ₈ OH-1	−24.76	84.18	26.75	38.48	50.25	55.54	63.98
iC ₄ H ₈ OH-2	−22.32	84.64	25.65	37.59	49.76	55.19	63.81
iC ₄ H ₈ OH-3	−18.02	86.82	25.65	37.59	49.76	55.19	63.81
iC ₄ H ₉ O	−14.76	81.94	25.44	37.89	50.54	56.13	64.88
tC ₄ H ₉ OH	−74.82	78.08	27.29	40.25	53.23	59.10	68.27
tC ₄ H ₉ OH ^a	−74.72	–	27.29	40.25	53.23	59.10	68.27
tC ₄ H ₈ OH	−23.02	83.68	26.52	38.34	50.07	55.36	63.61
tC ₄ H ₉ O	−21.76	76.62	26.31	38.64	50.85	56.30	64.68

^a The recommended values of NIST [32].

Table 3
Elementary reactions and rate constant expressions. Rate constants in the form, $A \exp(-E_a/RT)$, in cm^3 , mol, cal, and K. The Troe parameters listed in the order: a , T^* , T^{**} , T^{**} .

No.	Reaction	A	n	E_a	Ref.
(316)	$\text{C}_2\text{H}_3\text{OH} = \text{CH}_3\text{CHO}$	7.42×10^{46}	-10.56	67,420	[55]
(1663)	$\text{CH}_3\text{CHCHOH} = \text{C}_2\text{H}_5\text{CHO}$	8.59×10^{11}	0.30	55,900	[54]
(1664)	$\text{iC}_3\text{H}_5\text{OH} = \text{CH}_3\text{COCH}_3$	8.59×10^{11}	0.30	55,900	[54]
(1676)	$\text{iC}_3\text{H}_5\text{OH} + \dot{\text{H}} = \text{C}_2\text{H}_3\text{OH} + \dot{\text{C}}\text{H}_3$	6.26×10^{13}	0.00	4500	[57]
(1677)	$\text{iC}_3\text{H}_5\text{OH} + \dot{\text{H}} = \text{C}_3\text{H}_6 + \dot{\text{O}}\text{H}$	6.26×10^{13}	0.00	4500	[57]
(1678)	$\text{CH}_3\text{CHCHOH} + \dot{\text{H}} = \text{C}_2\text{H}_3\text{OH} + \dot{\text{C}}\text{H}_3$	6.26×10^{13}	0.00	4500	[57]
(1679)	$\text{CH}_3\text{CHCHOH} + \dot{\text{H}} = \text{C}_3\text{H}_6 + \dot{\text{O}}\text{H}$	6.26×10^{13}	0.00	4500	[57]
(1697)	$\text{C}_3\text{H}_6 + \dot{\text{O}}\text{H} = \text{iC}_3\text{H}_6\text{OH}$	2.92×10^{12}	0.00	-1000	[51]
(1698)	$\text{C}_2\text{H}_3\text{OH} + \dot{\text{C}}\text{H}_3 = \text{iC}_3\text{H}_6\text{OH}$	1.89×10^3	2.67	6850	[51]
(1699)	$\text{iC}_3\text{H}_5\text{OH} + \dot{\text{H}} = \text{iC}_3\text{H}_6\text{OH}$	6.25×10^{11}	0.50	2620	[51]
(1700)	$\text{C}_2\text{H}_5\text{CHO} + \dot{\text{H}} = \dot{\text{C}}_3\text{H}_6\text{OH-1}$	4.24×10^{11}	0.51	1230	[51]
(1701)	$\text{CH}_3\text{CHCHOH} + \dot{\text{H}} = \dot{\text{C}}_3\text{H}_6\text{OH-1}$	2.50×10^{11}	0.50	2620	[51]
(1702)	$\text{C}_2\text{H}_3\text{OH} + \dot{\text{C}}\text{H}_3 = \dot{\text{C}}_3\text{H}_6\text{OH-1}$	1.76×10^4	2.50	6130	[51]
(1703)	$\text{C}_3\text{H}_6 + \dot{\text{O}}\text{H} = \dot{\text{C}}_3\text{H}_6\text{OH-2}$	4.95×10^{11}	0.00	-960	[51]
(1704)	$\text{CH}_3\text{CHCHOH} + \dot{\text{H}} = \dot{\text{C}}_3\text{H}_6\text{OH-2}$	2.50×10^{11}	0.50	2620	[51]
(1705)	$\text{C}_3\text{H}_5\text{OH} + \dot{\text{H}} = \dot{\text{C}}_3\text{H}_6\text{OH-2}$	4.24×10^{11}	0.50	1230	[51]
(1719)	$\text{tC}_4\text{H}_9\text{OH} (+\text{M}) = \text{tC}_3\text{H}_6\text{OH} + \dot{\text{C}}\text{H}_3 (+\text{M})$ Troe parameters: 0.41, 1100, 490, 2850	$k_\infty 6.31 \times 10^{16}$ $k_0 9.82 \times 10^{74}$	0.00 -16.1	81,277 92,918	†
(1720)	$\text{tC}_4\text{H}_9\text{OH} (+\text{M}) = \text{tC}_4\text{H}_9 + \dot{\text{O}}\text{H} (+\text{M})$ Troe parameters: 0.37, 469, 2.91×10^{12} , 3670	$k_\infty 1.20 \times 10^{24}$ $k_0 4.53 \times 10^{74}$	-2.10 -16.3	97,281 107,810	†
(1721)	$\text{tC}_4\text{H}_9\text{OH} (+\text{M}) = \text{tC}_4\text{H}_9\dot{\text{O}} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.303, 516, 3.96×10^7 , 2390	$k_\infty 1.10 \times 10^{15}$ $k_0 5.44 \times 10^{61}$	0.00 -13.1	105,150 114,690	†
(1722)	$\text{tC}_4\text{H}_9\text{OH} (+\text{M}) = \text{tC}_4\text{H}_8\text{OH} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.321, 507, 8.00×10^6 , 2.54×10^3	$k_\infty 1.30 \times 10^{18}$ $k_0 1.54 \times 10^{65}$	-0.4 -13.7	104,460 114,110	†
(1723)	$\text{tC}_4\text{H}_9\text{OH} (+\text{M}) = \text{iC}_4\text{H}_8 + \text{H}_2\text{O} (+\text{M})$ Troe parameters: 3.27×10^{-3} , 333, 1.41×10^{10} , 3820	$k_\infty 1.48 \times 10^{14}$ $k_0 5.87 \times 10^{95}$	0.00 -22.6	64,200 79,149	†
(1724)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{iC}_3\text{H}_6\text{OH} + \dot{\text{C}}\text{H}_3 (+\text{M})$ Troe parameters: 4.25×10^{-2} , 278, 3.36×10^6 , 2190	$k_\infty 1.10 \times 10^{26}$ $k_0 1.53 \times 10^{84}$	-2.80 -19.1	92,354 104,670	†
(1725)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_2\text{H}_4\text{OH} + \dot{\text{C}}_2\text{H}_5 (+\text{M})$ Troe parameters: 5.58×10^{-3} , 358, 4.93×10^7 , 2480	$k_\infty 1.73 \times 10^{26}$ $k_0 1.67 \times 10^{89}$	-2.70 -20.3	86,207 99,453	†
(1726)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{C}_3\text{H}_6\text{OH-1} + \dot{\text{C}}\text{H}_3 (+\text{M})$ Troe parameters: 0.399, 217, 466, 2280	$k_\infty 8.50 \times 10^{24}$ $k_0 9.91 \times 10^{86}$	-2.50 -19.7	88,444 101,200	†
(1727)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_9 + \dot{\text{O}}\text{H} (+\text{M})$ Troe parameters: 8.68×10^{-5} , 293, 1.97×10^4 , 1670	$k_\infty 6.43 \times 10^{22}$ $k_0 1.90 \times 10^{79}$	-1.90 -17.5	96,229 108,350	†
(1728)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_8\text{OH-M} + \dot{\text{H}} (+\text{M})$ Troe parameters: 8.94×10^{-2} , 337, 7.84×10^9 , 1660	$k_\infty 1.33 \times 10^{16}$ $k_0 5.22 \times 10^{67}$	0.00 -14.3	102,430 113,840	†
(1729)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_8\text{OH-1} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.109, 86.8, 5.91×10^9 , 2440	$k_\infty 8.86 \times 10^{18}$ $k_0 6.42 \times 10^{75}$	-0.80 -16.8	94,238 106,460	†
(1730)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_8\text{OH-2} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.180, 274., 1.08×10^7 , 2150	$k_\infty 3.63 \times 10^{20}$ $k_0 2.71 \times 10^{71}$	-1.30 -15.7	102,100 113,570	†
(1731)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_8\text{OH-3} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.161, 286, 1.54×10^6 , 2060	$k_\infty 1.31 \times 10^{16}$ $k_0 1.74 \times 10^{67}$	0.00 -14.5	101,830 113,350	†
(1732)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{sC}_4\text{H}_9\dot{\text{O}} + \dot{\text{H}} (+\text{M})$ Troe parameters: 0.488, 27.2, 2.10×10^{10} , 2.25×10^6	$k_\infty 2.20 \times 10^{14}$ $k_0 8.60 \times 10^{62}$	0.20 -13.6	104,850 115,890	†
(1733)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{C}_4\text{H}_8-1 + \text{H}_2\text{O} (+\text{M})$ Troe parameters: 2.82×10^{-4} , 362, 1.29×10^4 , 4490	$k_\infty 5.00 \times 10^{13}$ $k_0 4.82 \times 10^{94}$	0.00 -22.5	67,960 83,529	†
(1734)	$\text{sC}_4\text{H}_9\text{OH} (+\text{M}) = \text{C}_4\text{H}_8-2 + \text{H}_2\text{O} (+\text{M})$ Troe parameters: 2.95×10^{-3} , 293, 5.21×10^{10} , 5080	$k_\infty 3.50 \times 10^{13}$ $k_0 4.63 \times 10^{101}$	0.00 -24.5	65,900 83,466	†
(1735)	$\text{iC}_4\text{H}_9\text{OH} (+\text{M}) = \text{iC}_3\text{H}_7 + \dot{\text{C}}\text{H}_2\text{OH} (+\text{M})$ Troe parameters: 0.134, 776, 4.01×10^7 , 4430	$k_\infty 3.25 \times 10^{25}$ $k_0 1.23 \times 10^{73}$	-2.40 -15.7	87,980 92,616	†
(1736)	$\text{iC}_4\text{H}_9\text{OH} (+\text{M}) = \dot{\text{C}}_3\text{H}_6\text{OH-2} + \dot{\text{C}}\text{H}_3 (+\text{M})$ Troe parameters: 0.144, 836, 4.84×10^6 , 4550	$k_\infty 4.00 \times 10^{27}$ $k_0 4.05 \times 10^{66}$	-3.10 -14.4	92,180 90,780	†
(1737)	$\text{iC}_4\text{H}_9\text{OH} (+\text{M}) = \text{iC}_4\text{H}_9 + \dot{\text{O}}\text{H} (+\text{M})$	$k_\infty 1.40 \times 10^{21}$	-1.50	94,890	†

Table 3 (continued)

No.	Reaction	A	n	E_a	Ref.
	Troe parameters: 5.45×10^{-4} , 1125, 1.68×10^{10} , 3635	$k_0 2.09 \times 10^{55}$	–11.2	90,431	
(1738)	$iC_4H_9OH(+M) = iC_4H_8OH-1 + H(+M)$ Troe parameters: 2.09×10^{-5} , 1160, 1.68×10^{11} , 3700	$k_\infty 5.68 \times 10^{16}$ $k_0 1.89 \times 10^{49}$	–0.30 –9.55	95,910 90,470	†
(1739)	$iC_4H_9OH(+M) = iC_4H_8OH-2 + H(+M)$ Troe parameters: 0.146, 1030, 1.07×10^{11} , 5130	$k_\infty 1.70 \times 10^{17}$ $k_0 2.48 \times 10^{46}$	–0.50 –8.81	98,200 90,900	†
(1740)	$iC_4H_9OH(+M) = iC_4H_8OH-3 + H(+M)$ Troe parameters: 0.271, 2740, 129, 9660	$k_\infty 5.00 \times 10^{17}$ $k_0 3.32 \times 10^{41}$	–0.50 –7.37	102,490 92,820	†
(1741)	$iC_4H_9OH(+M) = iC_4H_9O + H(+M)$ Troe parameters: 0.500, 6210, 666, 2.90×10^6	$k_\infty 1.20 \times 10^{15}$ $k_0 1.49 \times 10^{36}$	0.00 –6.14	105,180 94,570	†
(1742)	$iC_4H_9OH(+M) = iC_4H_8 + H_2O(+M)$ Troe parameters: 0.202, 1290, 1280, 6.79×10^4	$k_\infty 4.56 \times 10^{13}$ $k_0 6.10 \times 10^{101}$	0.00 –23.8	64,970 106,720	†
(1743)	$tC_4H_9OH + H = tC_4H_9O + H_2$	9.45×10^2	3.10	8701	[42]
(1744)	$tC_4H_9OH + H = tC_4H_8OH + H_2$	2.00×10^6	2.54	6756	[53]
(1745)	$tC_4H_9OH + OH = tC_4H_9O + H_2O$	5.88×10^2	2.82	–585	[45]
(1746)	$tC_4H_9OH + OH = tC_4H_8OH + H_2O$	6.93	3.70	–2416	[45]
(1751)	$tC_4H_9OH + CH_3 = tC_4H_9O + CH_4$	1.02	3.60	8221	†
(1752)	$tC_4H_9OH + CH_3 = tC_4H_8OH + CH_4$	1.36	3.60	9154	†
(1765)	$sC_4H_9OH + H = sC_4H_9O + H_2$	9.45×10^2	3.10	8701	[42]
(1766)	$sC_4H_9OH + H = sC_4H_8OH-M + H_2$	6.66×10^5	2.50	6756	[53]
(1767)	$sC_4H_9OH + H = sC_4H_8OH-3 + H_2$	6.66×10^5	2.50	6756	[53]
(1768)	$sC_4H_9OH + H = sC_4H_8OH-2 + H_2$	1.08×10^5	2.70	4440	†
(1769)	$sC_4H_9OH + H = sC_4H_8OH-1 + H_2$	4.40×10^4	2.70	2915	[42]
(1770)	$sC_4H_9OH + OH = sC_4H_9O + H_2O$	5.88×10^2	2.80	–585	†
(1771)	$sC_4H_9OH + OH = sC_4H_8OH-M + H_2O$	2.31	3.70	–2946	[45]
(1772)	$sC_4H_9OH + OH = sC_4H_8OH-3 + H_2O$	5.17×10^6	1.8	12	[45]
(1773)	$sC_4H_9OH + OH = sC_4H_8OH-2 + H_2O$	1.54	3.70	–3736	[45]
(1774)	$sC_4H_9OH + OH = sC_4H_8OH-1 + H_2O$	1.80×10^3	2.90	–2611	[45]
(1785)	$sC_4H_9OH + CH_3 = sC_4H_9O + CH_4$	1.02	3.60	8221	†
(1786)	$sC_4H_9OH + CH_3 = sC_4H_8OH-M + CH_4$	0.453	3.60	7154	†
(1787)	$sC_4H_9OH + CH_3 = sC_4H_8OH-3 + CH_4$	0.453	3.60	7154	†
(1788)	$sC_4H_9OH + CH_3 = sC_4H_8OH-2 + CH_4$	8.02	3.20	6461	†
(1789)	$sC_4H_9OH + CH_3 = sC_4H_8OH-1 + CH_4$	9.96	3.40	7634	†
(1820)	$iC_4H_9OH + H = iC_4H_9O + H_2$	9.45×10^2	3.10	8701	[42]
(1821)	$iC_4H_9OH + H = iC_4H_8OH-3 + H_2$	1.33×10^6	2.50	6756	[53]
(1822)	$iC_4H_9OH + H = iC_4H_8OH-2 + H_2$	6.50×10^5	2.40	4471	[53]
(1823)	$iC_4H_9OH + H = iC_4H_8OH-1 + H_2$	8.79×10^4	2.70	2915	[42]
(1824)	$iC_4H_9OH + OH = iC_4H_9O + H_2O$	5.88×10^2	2.80	–585	[45]
(1825)	$iC_4H_9OH + OH = iC_4H_8OH-3 + H_2O$	7.67×10^6	1.8	–147	[45]
(1826)	$iC_4H_9OH + OH = iC_4H_8OH-2 + H_2O$	0.77	3.70	–4940	[45]
(1827)	$iC_4H_9OH + OH = iC_4H_8OH-1 + H_2O$	3.61×10^3	2.90	–2331	[45]
(1836)	$iC_4H_9OH + CH_3 = iC_4H_9O + CH_4$	1.02	3.60	8221	†
(1837)	$iC_4H_9OH + CH_3 = iC_4H_8OH-3 + CH_4$	0.960	3.60	7154	†
(1838)	$iC_4H_9OH + CH_3 = iC_4H_8OH-2 + CH_4$	0.755	3.50	5481	†
(1839)	$iC_4H_9OH + CH_3 = iC_4H_8OH-1 + CH_4$	19.9	3.40	7634	†
(1864)	$iC_4H_8 + OH = tC_4H_8OH$	9.93×10^{11}	0.00	–960	[10]
(1865)	$iC_3H_5OH + CH_3 = tC_4H_8OH$	1.30×10^3	2.50	8520	[51]
(1866)	$CH_3CHCHOH + CH_3 = iC_4H_8OH-1$	1.89×10^3	2.70	6850	[51]
(1867)	$iC_3H_6CHOH + H = iC_4H_8OH-1$	6.25×10^{11}	0.50	2620	[51]
(1868)	$iC_3H_7CHO + H = iC_4H_8OH-1$	8.00×10^{12}	0.00	9500	[52]
(1869)	$iC_3H_5CH_2OH + H = iC_4H_8OH-2$	1.06×10^{12}	0.50	1230	[51]
(1870)	$iC_3H_6CHOH + H = iC_4H_8OH-2$	2.50×10^{11}	0.50	2620	[51]
(1871)	$iC_4H_8 + OH = iC_4H_8OH-2$	9.93×10^{11}	0.00	–960	[10]
(1872)	$C_3H_6 + CH_2OH = iC_4H_8OH-3$	9.45×10^2	2.70	6850	[51]
(1873)	$C_3H_5OH + CH_3 = iC_4H_8OH-3$	1.89×10^3	2.70	6850	[51]
(1874)	$iC_3H_5CH_2OH + H = iC_4H_8OH-3$	6.25×10^{11}	0.50	2620	[51]
(1875)	$C_4H_7OH1-2 + H = sC_4H_8OH-M$	2.50×10^{11}	0.50	2620	[51]
(1876)	$C_4H_8-1 + OH = sC_4H_8OH-M$	9.93×10^{11}	0.00	–960	[10]
(1877)	$\dot{C}_2H_5 + C_2H_3OH = sC_4H_8OH-M$	8.80×10^3	2.50	6130	[51]
(1878)	$C_4H_7OH1-2 + H = sC_4H_8OH-1$	4.24×10^{11}	0.50	1230	[51]
(1879)	$C_2H_5COCH_3 + H = sC_4H_8OH-1$	8.00×10^{12}	0.00	9500	[52]

(continued on next page)

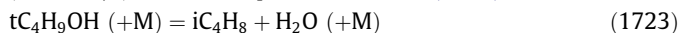
Table 3 (continued)

No.	Reaction	A	n	E _a	Ref.
(1880)	C ₄ H ₇ OH ₂ -2 + H = sC ₄ H ₈ OH-1	2.50 × 10 ¹¹	0.50	2620	[51]
(1881)	iC ₃ H ₅ OH + CH ₃ = sC ₄ H ₈ OH-1	1.76 × 10 ⁴	2.50	6130	[51]
(1882)	CH ₃ CHCHOH + CH ₃ = sC ₄ H ₈ OH-2	1.89 × 10 ³	2.70	6850	[51]
(1883)	C ₄ H ₇ OH ₂ -2 + H = sC ₄ H ₈ OH-2	2.50 × 10 ¹¹	0.50	2620	[51]
(1884)	C ₄ H ₈ -2 + OH = sC ₄ H ₈ OH-2	9.93 × 10 ¹¹	0.00	−960	[10]
(1885)	C ₄ H ₇ OH ₁ -3 + H = sC ₄ H ₈ OH-2	4.24 × 10 ¹¹	0.50	1230	[51]
(1886)	C ₄ H ₇ OH ₁ -3 + H = sC ₄ H ₈ OH-3	2.50 × 10 ¹¹	0.50	2620	[51]
(1887)	sC ₂ H ₄ OH + C ₂ H ₄ = sC ₄ H ₈ OH-3	9.45 × 10 ²	2.70	6850	[51]

† This work.

were calculated based on microscopic reversibility using an estimate of the rate constant in the reverse direction for radical–radical recombination. For reaction (−1720) a rate constant of $2.41 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [37] for recombination of hydroxyl and isobutyl radicals to form isobutanol. For both of the reactions (−1721) and (−1722) a rate constant of $1.0 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used for hydrogen atom addition to another free radical.

A high-pressure limit rate constant of $1.48 \times 10^{14} \exp(-32,300/T) \text{ s}^{-1}$ was adopted for reaction (1723):



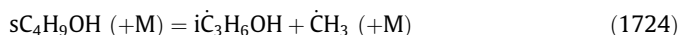
This value was taken from the unimolecular water elimination reaction of 2,3-DMB-2-ol measured by Tsang [36].

A chemical activation formulation based on Quantum Rice–Ramsperger–Kassel theory, as described by Dean [38,39] was then used together with the high-pressure limit expressions to develop pressure-dependent rate constants which were fitted to a nine-parameter Troe formalism [40].

3.1.2. *sec*-Butanol

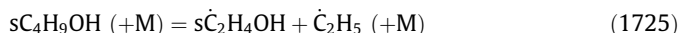
As before, the *sec*-butanol sub-mechanism includes the two possible classes of unimolecular decomposition reaction.

The high-pressure limit rate constants for each of these reactions were calculated based on microscopic reversibility using an estimate of the rate constant in the reverse direction for radical–radical recombination. For reaction (−1724):



a rate constant of $1.93 \times 10^{14} T^{-0.32} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [37] for the recombination of methyl and isobutyl radicals to form isopentane.

For reaction (−1725):



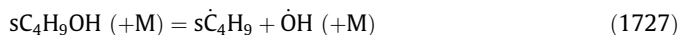
a rate constant of $1.15 \times 10^{14} T^{-0.35} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [41] for recombination of ethyl and isopropyl radicals to form isopentane. In the decomposition of *sec*-butanol the rate of reaction (1725) is very important. The estimated value initially adopted is far too slow to explain the consumption of the concentration of *sec*-butanol measured in the pyrolysis work, so it was increased by a factor of three before being used in the simulations.

For reaction (−1726):



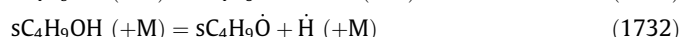
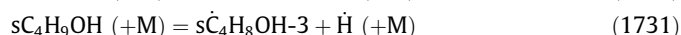
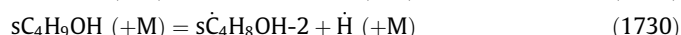
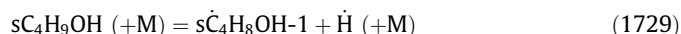
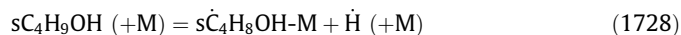
a rate constant of $1.36 \times 10^{15} T^{-0.68} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [41] for recombination of methyl and isopropyl radicals to form isobutane.

For reaction (−1727):



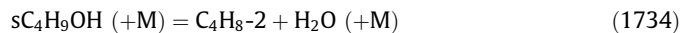
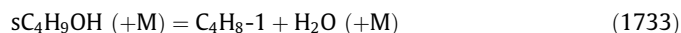
a rate constant of $2.41 \times 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [37] for the recombination of hydroxyl and isobutyl radicals to form isobutanol.

For reactions (−1728) to (−1732):



a rate constant of $1.0 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used for hydrogen atom addition to a larger radical.

A high-pressure limit rate constant of $5.0 \times 10^{13} \exp(-34,200/T) \text{ s}^{-1}$ was adopted for reaction (1733):

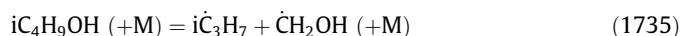


This was half the value estimated by Tsang [36] as a maximum value of unimolecular water elimination reaction of 3,3-DMB-2-ol.

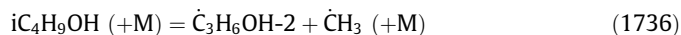
Considering the difference of hydrogen atoms positioned at primary or secondary carbon atoms for reactions (1733) and (1734), the high-pressure limit rate constant of $3.5 \times 10^{13} \exp(-33200/T) \text{ s}^{-1}$ was adopted for reaction (1734). Pressure-dependent rate constants were estimated by the same method adopted in the case of *tert*-butanol.

3.1.3. *Isobutanol*

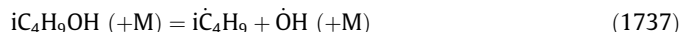
The isobutanol sub-mechanism includes, as before, the two possible classes of unimolecular decomposition reaction, including simple bond scission and water elimination reactions. For reaction (−1735):



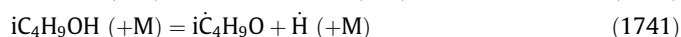
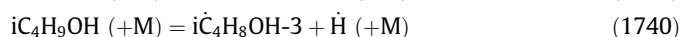
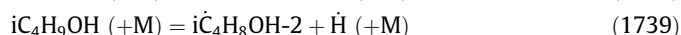
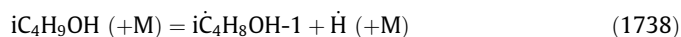
a rate constant of $1.15 \times 10^{14} T^{-0.35} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used as recommended by Tsang [37] for recombination of ethyl and isopropyl radicals to form isopentane. For reaction (−1736):



a rate constant of $1.15 \times 10^{14} T^{-0.35} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used based on the recombination of methyl and isobutyl radicals to form isopentane. For reaction (−1737):

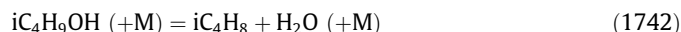


a rate constant of $1.93 \times 10^{14} T^{-0.32} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used based on the recombination of hydroxyl and isobutyl radicals to form isobutanol. For reactions (−1738) to (−1741):



a rate constant of $1.0 \times 10^{14} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ was used for hydrogen atom addition to a larger free radical.

A high-pressure limit rate constant of $4.58 \times 10^{14} \exp(-32,700/T) \text{ s}^{-1}$ was adopted for reaction (1742):



This value was taken from the unimolecular water elimination reaction of 2,3-DMB-2-ol as described by Tsang [36]. Pressure-dependent rate constants were estimated by the same method as adopted for *tert*- and *sec*-butanol.

3.2. Hydrogen abstraction

Rate constants for hydrogen abstraction by $\dot{H}$, $\dot{CH}_3$, $\dot{OH}$ radicals and other atoms and radicals from the α , β and γ positions of butanol isomers were considered in this study. The rate constants were assumed to be the same with those for *n*-butanol used by Black et al. [10] except the abstraction rates by $\dot{H}$ atoms and $\dot{OH}$ radicals. *n*-Butanol has hydrogen positioned on α , β , γ and δ carbons [33–35], whose bond dissociation energies are 95.5(α), 100.6(β), 98.8(γ) and 101.9(δ) kcal mol⁻¹, respectively. The H atom abstraction rates by H atoms from α carbon were updated based on the ethanol + H work of Sivaramakrishnan et al. [42]. The H atom abstraction rates by $\dot{OH}$ radicals were estimated based on ab initio calculations by Zhou et al. [43] and Zador et al. [44]. These rates are examined in the parallel work performed by Sarathy et al. [45] which includes low temperature studies.

sec-Butanol has hydrogen positioned at α , two different β , and γ , of which bond dissociation energies are 94.7 (α), 102.6 (β) on methyl group, 100.8 (β) on ethyl group and 102.0 (γ) kcal mol⁻¹, respectively. The rate constants for hydrogen abstraction from α of *sec*-butanol were taken from α of *n*-butanol. Those from β on methyl group, β on ethyl group and γ were taken from δ , β and (δ), respectively. Isobutanol has hydrogen positioned at α , β and γ , of which bond dissociation energies are 95.9 (α), 97.6 (β) and 101.8 (γ) kcal mol⁻¹, respectively. The rate constants for hydrogen abstraction from α , β and γ of isobutanol were taken from α , γ and δ of *n*-butanol, respectively.

tert-Butanol has hydrogen only on β carbon atoms, whose bond dissociation energies are 103.9 kcal mol⁻¹. The rate constant for hydrogen abstraction from the β -carbons of *tert*-butanol were assumed equal to abstraction from δ -carbons of *n*-butanol. All of these bond dissociation energies were computed using the CBS-QB3 methodology [46,47] as implemented in Gaussian-03 [48] together with isodesmic [49] and isogeitonic [50] reactions.

3.3. Radical decomposition

The radicals listed in Table 3 are produced through the pyrolysis and oxidation of butanol isomers. These radicals mainly decompose via β -scission at temperatures above approximately 1000 K. We considered unimolecular reactions of radicals in the reverse, exothermic direction, as the addition of an alkyl radical to a C=O and a C=C moiety. These rate constants were estimated based on the recommendation of Curran [51] and estimation for the hydrogen atom addition to acetaldehyde by Natarajan et al. [52].

Reactions (–1864), (–1871), (–1876) and (–1884):



were considered to be similar to a hydroxyl radical addition to 1-butene with a rate constant of $9.93 \times 10^{11} \exp(+483/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for hydroxyl radical addition to 1-butene, 2-butene and isobutene. This value was taken from Black et al. [10]. Reaction (–1865) was considered to be similar to a methyl radical

addition to isobutene with a rate constant of $1.30 \times 10^{10} T^{2.48} \exp(-4,287/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.

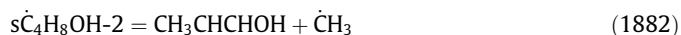
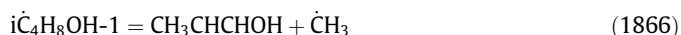


Reactions (–1872), (–1877) and (–1887):



were considered to be similar to ethyl radical addition to propene with rate constants of $8.80 \times 10^{13} T^{2.48} \exp(-3,071/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $9.45 \times 10^{13} T^{2.67} \exp(-3,447/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$, which are half the value of pre-exponential factor of methyl addition to propene.

Reactions (–1866), (–1873), (–1881) and (–1882):



were treated as being similar to methyl radical addition to propene with a rate constant of $1.89 \times 10^{13} T^{2.67} \exp(-3,447/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.

Reactions (–1869), (–1870), (–1875), (–1878), (–1880), (–1883), (–1885) and (–1886):



were considered to be similar to hydrogen atom addition to propene with rate constants of $2.50 \times 10^{11} T^{0.51} \exp(-1,318/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $4.24 \times 10^{11} T^{0.51} \exp(-619/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.

Reactions (–1868) and (–1879)



were considered to be similar to hydrogen atom addition to acetaldehyde with a rate constant of $8.50 \times 10^{12} \exp(-4,780/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$.

Reactions (–1867) and (–1874):



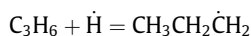
were considered to be similar to hydrogen atom addition to isobutene with rate constants of $6.25 \times 10^{11} T^{0.51} \exp(-1,318/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for addition to the internal carbon atom and

$1.06 \times 10^{12} T^{0.51} \exp(-619/T) \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ for addition to the terminal carbon atom.

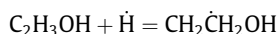
3.4. Enol reactions

Under conditions of high temperature pyrolysis and oxidation, significant amount of radicals are generally produced. Radicals mainly decompose through β -scission to produce species with unsaturated bonds above 1000 K. The chemistry of unsaturated alcohols or enols should be important in butanol isomers pyrolysis and oxidation. It is well known that enol isomerizes to aldehyde or ketone very easily in the condensed phase, the so called keto-enol tautomerization; however in the gas-phase enols are now well known though their reactivity has been little explored experimentally. The rate constant of tautomerization of vinyl alcohol to acetaldehyde was calculated by da Silva et al. [54] at the CBS-APNO level of theory as $8.59 \times 10^{11} T^{0.318} \exp(-28,132/T) \text{ s}^{-1}$, which is adopted as the rate constant of enol to aldehyde or ketone isomerization except vinyl alcohol in this work. The pressure dependent rate constant for the tautomerization of vinyl alcohol was adopted based on the theoretical work of Shao et al. [55].

Hydrogen atom addition to unsaturated bonds should also be important in enol chemistry. Simmie et al. [56] investigated the energy barrier of H addition to propene:



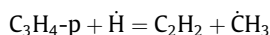
and vinyl alcohol:



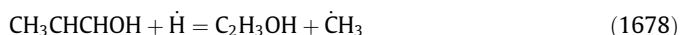
reporting values of 2.3 and 3.7 kcal mol⁻¹, respectively.

The products, $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{H}_2$ and $\text{CH}_2\dot{\text{C}}\text{H}_2\text{OH}$, can decompose to methyl and hydroxyl radicals. Judging from the barrier difference of 1.4 kcal mol⁻¹ between the reactions mentioned above, methyl elimination reaction preceded by hydrogen atom addition seems to be slightly predominant, however, we considered both of the methyl and hydroxyl radicals elimination reactions.

From the analogy of the following hydrogen atom addition reaction to propyne:



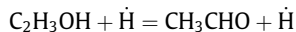
the rate constant proposed by Rosado-Reyes et al. [57] was adopted as hydrogen atom addition to enols, for example:



Hydrogen atom can catalyze the hydrogen shift that converts an enol into aldehydes or ketones. For example, hydrogen atom can add to the vinyl alcohol to form a chemically activated $\text{sC}_2\text{H}_4\text{OH}^*$ radical, which can subsequently decompose to produce a hydrogen atom and acetaldehyde. This reaction is included in the following form in our reaction mechanism:



The hydrogen shift reactions of enols catalyzed by hydrogen atom proceed via the chemically activated radicals. The specific reactions like,



must be adopted.

Unfortunately we could not distinguish the concentration of enols from their corresponding species in this study, so these reactions were not included.

It was theoretically revealed by da Silva et al. [58] that HO_2 can also catalyze the hydrogen shift of enols, and this reaction will be discussed by Sarathy et al. [45].

4. Results and discussion

Simulations were performed with CHEMKIN-PRO [59], using the Closed Homogeneous Batch Reactor model at constant volume for all our shock tube experiments. As an integral part of the simulations, sensitivity analyses were carried out by increasing and decreasing both the forward and reverse rate constants, k , by a factor of two, with sensitivities expressed using the formula:

$$S = \frac{\log(X_+/X_-)}{\log(k_+/k_-)} = \frac{\log(X_+/X_-)}{\log(2/0.5)}$$

where X denotes the concentration of a particular species. For example, the sensitivity coefficients of species concentrations for a 1.5% n -butanol pyrolysis at 1300 K, 2.0 atm and 2.6 ms are shown in Fig. 3. A positive sensitivity indicates an increase in the species concentrations, conversely a negative value indicates a decrease in the species concentrations. Only those reactions which have sensitivity coefficients over 0.1 as absolute values are shown. The primary n -butanol reactions, in particular C–C bond scission reactions, have negative sensitivities to n -butanol concentration.

4.1. Pyrolysis

Enols cannot be detected by the analytical techniques employed in this work because surface-catalyzed reactions rapidly convert enols into aldehydes (or ketones). In order to compare measured and simulated concentrations we therefore summed the enol and aldehyde concentrations emerging from the simulations and matched these against the measured values.

4.1.1. n -Butanol

Major species concentration profiles were simulated for a mixture of 1.5% n -butanol diluted with argon in the pressure range 1.2–2.2 atm with effective heating times of 2.4–3.2 ms. The result is shown in Fig. 2a.

The observed concentrations of n -butanol have large error bars due to the strong adsorption. The concentration of n -butanol simply decreases as the temperature rises. The concentrations of ethylene, acetylene, ethane, carbon monoxide, methane and allene increase monotonically as the temperature increases. The concentration of acetaldehyde increases up to ≈ 1300 K and then decreases. The concentration of 1-butene increases to approximately 1250 K, then decreases. The model predicts these species concentration versus temperature profiles well.

Acetaldehyde is formed as one of the major products of n -butanol pyrolysis, Fig. 2a. Thus, oxy-containing compounds are produced as stable and intermediate species. We focus on the time-dependent behavior of the butanol isomers, aldehydes, ketones and enols in this study. The simulated time-dependent concentration profiles of n -butanol, vinyl alcohol ($\text{CH}_2=\text{CHOH}$) and acetaldehyde are shown in Fig. 2b. n -Butanol simply decomposes and 20% is consumed at approximately 75 μs . The concentration of acetaldehyde simply increases as the reaction proceeds. The concentration of vinyl alcohol increases to ≈ 360 μs and then declines. In order to explain this behavior, a reaction path analysis was carried out for the shock tube conditions outlined in Fig. 2c.

The reaction scheme shows that n -butanol undergoes a four-centered elimination to water and 1-butene, simple C–C bond scission reactions to produce methyl and $\dot{\text{C}}_3\text{H}_6\text{OH}$ radicals, ethyl and methoxymethyl ($\text{p}\dot{\text{C}}_2\text{H}_4\text{OH}$) radicals, propyl and hydroxymethyl

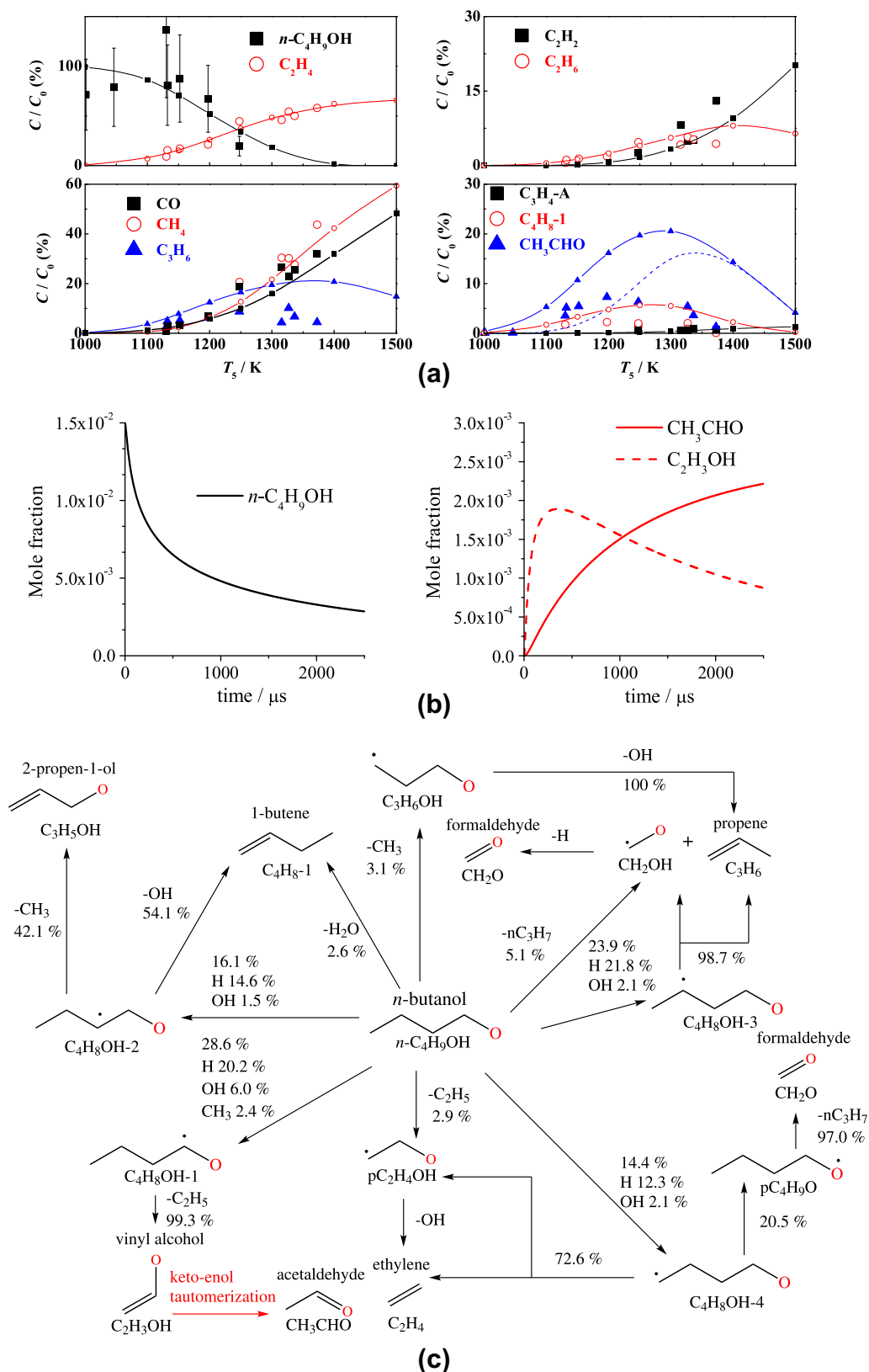


Fig. 2. (a) Normalized species concentration profiles for the pyrolysis of 1.5% $n\text{-butanol}$ diluted in Ar, effective heating time = 2.6 – 3.0 ms, lines simulation. C_0 and C denote the initial concentrations of $n\text{-C}_4\text{H}_9\text{OH}$ and products, respectively. The dashed line for acetaldehyde stands for the concentration of acetaldehyde, while solid line represents the summation of acetaldehyde and vinyl alcohol. (b) Simulated concentration profiles for 1.5% $n\text{-C}_4\text{H}_9\text{OH}$ diluted in Ar at 1300 K and 1.9 atm. (c) Reaction path analysis for $n\text{-C}_4\text{H}_9\text{OH}$ pyrolysis. Shock conditions: 1.5% $n\text{-C}_4\text{H}_9\text{OH}$ diluted in Ar, 1300 K, 1.9 atm, 20% consumption.

($\dot{\text{C}}_2\text{H}_3\text{OH}$) radicals, and hydrogen atom abstraction producing $\dot{\text{C}}_4\text{H}_8\text{OH-1}$, $\dot{\text{C}}_4\text{H}_8\text{OH-2}$, $\dot{\text{C}}_4\text{H}_8\text{OH-3}$ and $\dot{\text{C}}_4\text{H}_8\text{OH-4}$ radicals, which

accounts for 2.6%, 3.1%, 2.9%, 5.1%, 28.6%, 16.1%, 23.9% and 14.4% of total $n\text{-butanol}$ consumed, respectively.

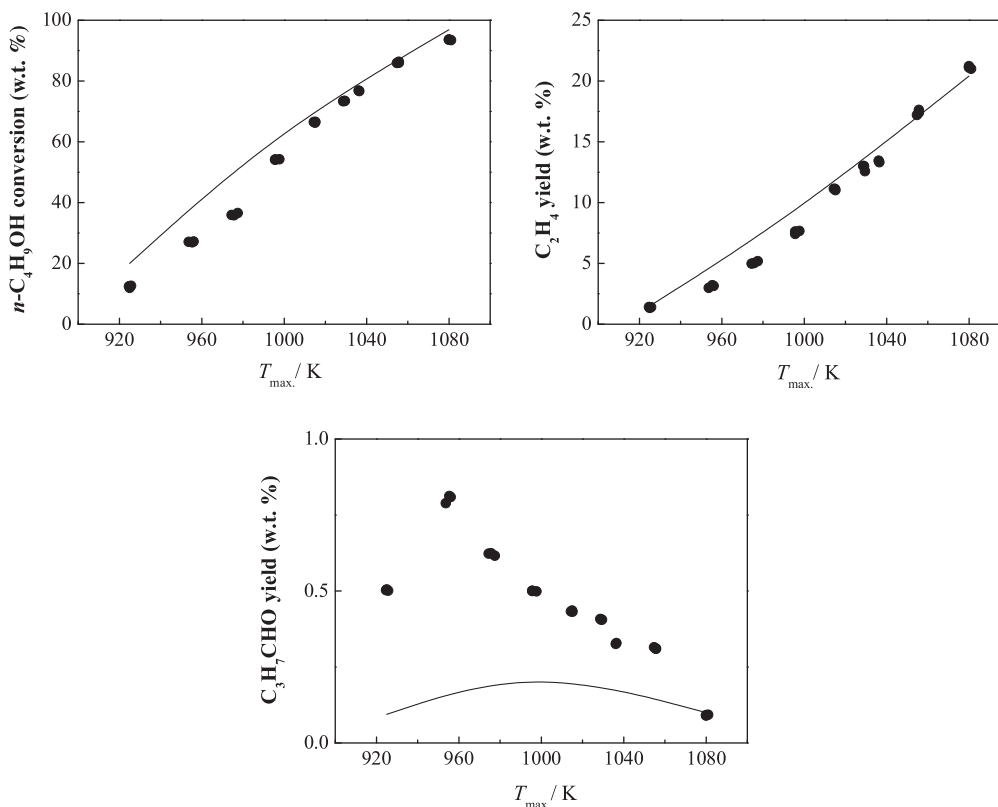


Fig. 4. Conversion and yield profiles for $n\text{-C}_4\text{H}_9\text{OH}$, ethylene and $n\text{-butyraldehyde}$ ($\text{C}_3\text{H}_7\text{CHO}$) in the pyrolysis reactor at 1.7 atm measured by Harper et al. [21]. The simulation line for $n\text{-butyraldehyde}$ represents the summation of $n\text{-butyraldehyde}$ and 1-buten-1-ol.

The reaction scheme shows that *sec*-butanol undergoes a four-centered elimination to water and 1-butene, or water and 2-butene, simple C–C bond scission to produce ethyl and $\text{s}\dot{\text{C}}_2\text{H}_4\text{OH}$ radicals, and hydrogen atom abstraction producing $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-1}$, $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$, $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-3}$ and $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-M}$ radicals, which accounts for 5.6%, 8.4%, 21.6%, 15.3%, 16.5%, 13.3% and 13.5% of total *sec*-butanol consumed, respectively.

The $\text{s}\dot{\text{C}}_2\text{H}_4\text{OH}$ radical produced from simple C–C bond scission, undergoes β -scission to form acetaldehyde and a hydrogen atom, or vinyl alcohol and a hydrogen atom, respectively. $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-1}$, $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$, $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-3}$ and $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-M}$ radicals produced from hydrogen atom abstraction also undergo β -scission. The $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-1}$ radical decomposes to produce a methyl radical and propen-2-ol ($\text{iC}_3\text{H}_5\text{OH}$), the latter subsequently isomerizing to acetone. $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$ decomposes to produce a hydroxyl radical and an isobutene, or a methyl radical and 1-propenol (CH_3CHCHOH). The 1-propenol can isomerize to propionaldehyde or react with a hydrogen atom to produce a methyl radical and vinyl alcohol, or a hydroxyl radical and propene. The $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-3}$ radical decomposes to produce ethylene and a $\text{s}\dot{\text{C}}_2\text{H}_4\text{OH}$ radical. The $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-M}$ radical decomposes to produce a hydroxyl radical and isobutene, or ethylene and vinyl alcohol – which subsequently isomerizes to acetaldehyde.

Sensitivity analyses were also carried out for the pyrolysis of *sec*-butanol. Figure 8 depicts the sensitivity coefficients of species concentrations for a 1.5% *sec*-butanol pyrolysis at 1300 K, 1.9 atm and 2.6 ms. The C–C bond scission of *sec*-butanol has strong negative sensitivity to *sec*-butanol concentration. Hydrogen atom abstraction from *sec*-butanol by hydrogen atom to produce $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-1}$ radical has positive sensitivity to the concentrations of acetone and propen-2-ol, because $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-1}$ mainly decomposes to produce a propen-2-ol and a methyl radical, consecutively propen-2-ol isomerizes to acetone. The isomerization reaction of

propen-2-ol to acetone has negative and positive sensitivities to the concentration of propen-2-ol and acetone. The branching ratio of $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$ decomposition and hydrogen abstraction from *sec*-butanol by hydrogen atom to produce $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$ have strong sensitivity to the concentration propionaldehyde and 1-propenol, because 1-propenol is produced by the C–C bond scission of $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-2}$ radical, which has two major decomposition paths, C–C and C–O bond scission. 1-Propenol consecutively isomerizes to propionaldehyde. The C–C bond scission of *sec*-butanol to produce an ethyl and a $\text{s}\dot{\text{C}}_2\text{H}_4\text{OH}$ radical and hydrogen abstraction from *sec*-butanol by hydrogen atom to produce $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-3}$ or $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-M}$ have strong sensitivity to the concentration of vinyl alcohol, because it is produced by the C–H bond scission of $\text{s}\dot{\text{C}}_2\text{H}_4\text{OH}$ or the C–C bond scission of $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-3}$ and $\text{s}\dot{\text{C}}_4\text{H}_8\text{OH-M}$ radical. The isomerization reaction of vinyl alcohol to acetaldehyde has negative and positive sensitivities to the concentration of vinyl alcohol and acetaldehyde.

sec-Butanol conversion and products yield profiles measured by Van Geem et al. [20] were also simulated for *sec*-butanol pyrolysis at the pressure 1.7 atm. The result is shown in Fig. 9. The *sec*-butanol conversion and ethylene, acetone and CO yield simply increase as temperature rises. The model predicts these species concentration versus temperature profiles well.

4.1.4. Isobutanol

Major species concentration profiles were simulated for a mixture of 1.5% isobutanol diluted with argon in the pressure range 1.2–2.2 atm with effective heating times in the range 2.6–3.0 ms. The result is shown in Fig. 10a. The concentration of isobutanol simply decreases as temperature rises. The concentrations of ethylene, acetylene, ethane, carbon monoxide, allene and methane simply increase as the temperature rises under our experimental conditions. The concentrations of propene and acetaldehyde

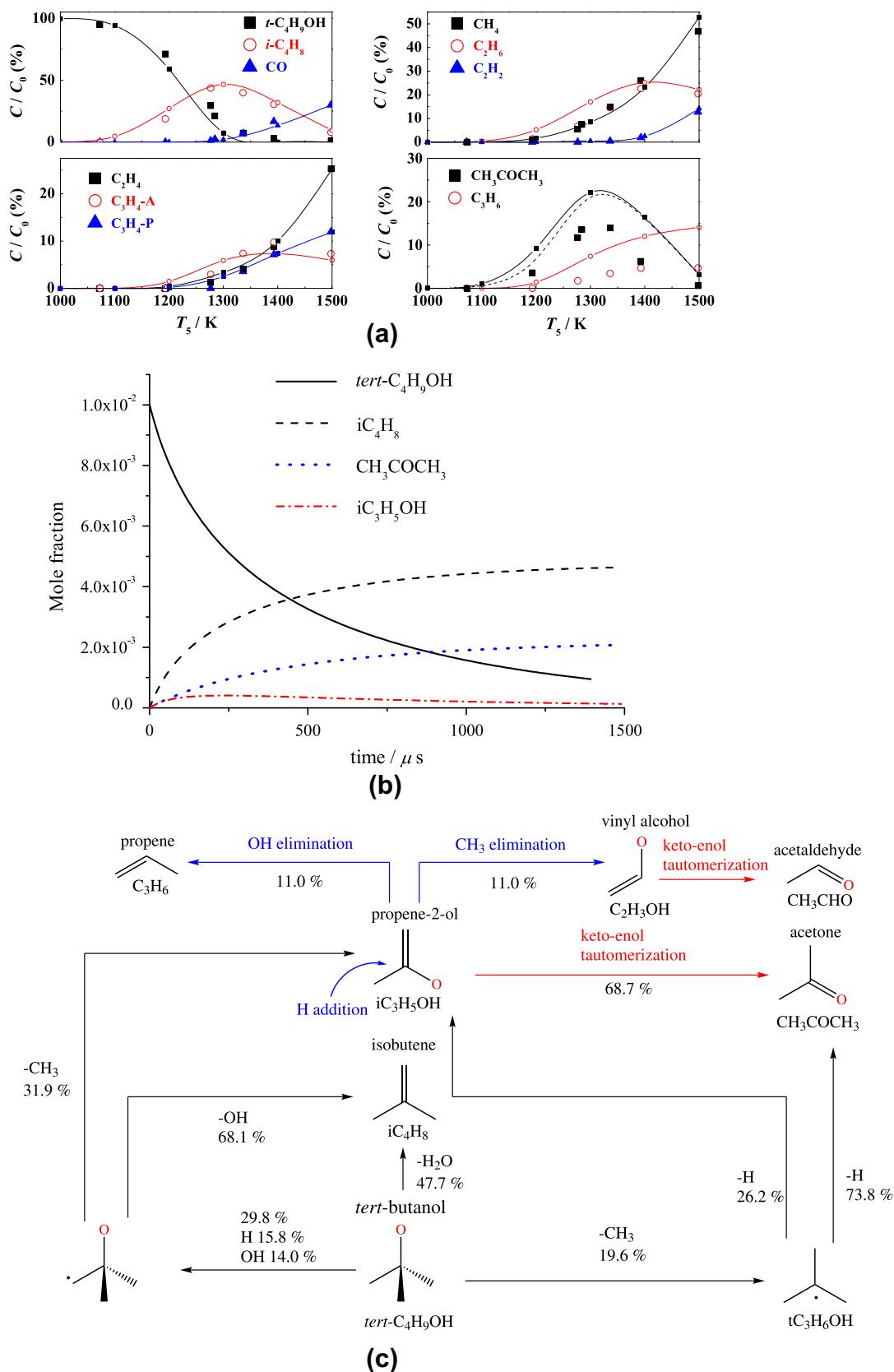


Fig. 5. (a) Normalized species concentration profiles for the pyrolysis of 1.0% $tert$ -C₄H₉OH in Ar, effective heating time = 1.5 – 1.7 ms, lines simulation. C_0 and C denote the initial concentrations of $tert$ -C₄H₉OH and products, respectively. The dashed line for acetone stands for the concentration of acetone, while solid line represents the summation of acetone and propene-2-ol. (b) Simulated concentration profiles for 1.0% $tert$ -C₄H₉OH diluted in Ar at 1300 K and 1.7 atm. (c) Reaction path analysis for $tert$ -C₄H₉OH pyrolysis. Shock conditions: 1.0% $tert$ -C₄H₉OH diluted in Ar, 1300 K, 1.7 atm, 20 % consumption.

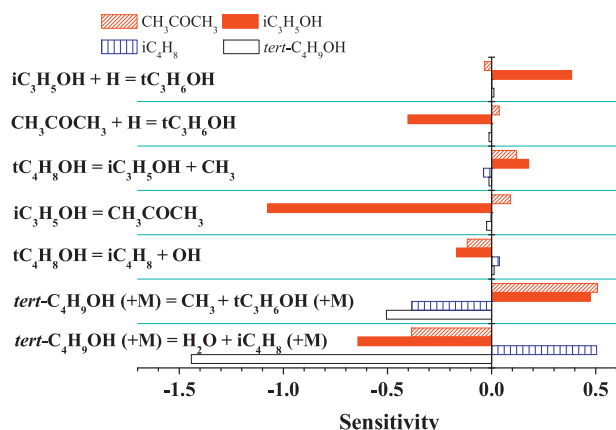


Fig. 6. Sensitivity to species concentration; 1.0% *tert*-C₄H₉OH diluted in Ar, 1300 K, 1.7 atm, $t_e = 1.6$ ms.

increase up to ≈ 1300 K, then decrease. The concentrations of isobutene and propionaldehyde increase up to ≈ 1250 K, then decrease. The model captures quite well these species versus temperature profiles except for the concentration of methane.

The simulated time-dependent concentration profiles of isobutanol, propionaldehyde, propen-1-ol (CH_3CHCHOH), propen-2-ol ($\text{iC}_3\text{H}_5\text{OH}$), vinyl alcohol ($\text{CH}_2=\text{CHOH}$) and acetaldehyde are shown in Fig. 10b. Isobutanol simply decomposes and 20% is consumed at ≈ 150 μs . The concentrations of 1-propenol, propionaldehyde and acetaldehyde simply increase as the reaction proceeds. The concentrations of vinyl alcohol and 1-propenol increase to approximately 1300 and 75 μs , then decrease. In order to explain these behaviors, a reaction path analysis was carried out for the shock tube conditions outlined in Fig. 10c. The reaction scheme shows that the isobutanol undergoes four-centered elimination to water and isobutene, simple C–C bond scission reactions to produce an isopropyl and a formyl radical, and hydrogen atom abstraction producing $\text{iC}_4\text{H}_8\text{OH-1}$, $\text{iC}_4\text{H}_8\text{OH-2}$ and $\text{iC}_4\text{H}_8\text{OH-3}$ radicals, which accounts for 13.0%, 13.7%, 29.5%, 11.6% and 28.3% of total isobutanol consumed, respectively.

The isopropyl and $\dot{\text{C}}_2\text{H}_2\text{OH}$ radicals produced from simple C–C bond scission undergo β -scission to form propene and a hydrogen atom, formaldehyde and a hydrogen atom, respectively. The $\text{C}_4\text{H}_8\text{OH-1}$, $\text{iC}_4\text{H}_8\text{OH-2}$ and $\text{iC}_4\text{H}_8\text{OH-3}$ radicals produced from hydrogen atom abstraction also undergo β -scission. $\text{iC}_4\text{H}_8\text{OH-1}$ radical has two major decomposition pathways. One produces a hydrogen atom and 2-methylpropen-1-ol ($\text{CH}_3\text{C}(\text{CH}_3)\text{CHOH}$ or $\text{iC}_3\text{H}_6\text{CHOH}$), while the other forms a methyl radical and 1-propenol, 1-propenol subsequently isomerizes to propionaldehyde or reacts with hydrogen atom to produce vinyl alcohol and methyl radical. The $\text{iC}_4\text{H}_8\text{OH-2}$ radical mainly decomposes to produce hydroxyl radical and isobutene. The $\text{iC}_4\text{H}_8\text{OH-3}$ radical has two main decomposition pathways; one produces a methyl radical and 1-propenol, while the other produces propene and a $\dot{\text{C}}_2\text{H}_2\text{OH}$ radical.

Sensitivity analyses were also carried out for the pyrolysis of isobutanol. Figure 11 depicts the sensitivity coefficients of species concentrations for a 1.5% isobutanol pyrolysis at 1300 K, 1.9 atm and 2.6 ms. The unimolecular decomposition reactions of isobutanol have negative sensitivity to isobutanol concentration. The hydrogen atom abstraction reactions from isobutanol by hydrogen atom, methyl and hydroxyl radical to produce $\text{iC}_4\text{H}_8\text{OH-1}$ have positive sensitivity to the concentrations of propionaldehyde and 1-propenol, because $\text{iC}_4\text{H}_8\text{OH-1}$ radical mainly decomposes to form a methyl radical and 1-propenol, which isomerizes to propionaldehyde. The C–C bond scission reaction of isobutanol and hydrogen atom abstraction from isobutanol by hydrogen atom, methyl and hydroxyl radical to produce $\text{iC}_4\text{H}_8\text{OH-1}$ radical have positive sensi-

titivities to the concentration of acetaldehyde and vinyl alcohol, conversely the isomerization reaction of 1-propenol to propionaldehyde has negative sensitivity. 1-Propenol produced by the decomposition of $\text{iC}_4\text{H}_8\text{OH-1}$ radical reacts with hydrogen atom to produce methyl radical and vinyl alcohol, subsequently vinyl alcohol isomerizes to acetaldehyde. This reaction competes with the isomerization of 1-propenol to propionaldehyde.

4.1.5. Ketones and aldehydes

Acetone, acetaldehyde and propionaldehyde were detected in the experiments of pyrolysis of butanol isomers. The simulation results qualitatively capture the behavior of those concentrations well, however, the discrepancies of quantity were observed for aldehydes and acetone. We discuss the case of propionaldehyde produced by the pyrolysis of *sec*-butanol, which is shown in Fig. 7a. The simulation underestimates the experimental data. Propionaldehyde is produced by the tautomerization of 1-propenol as shown in Fig. 7c. 1-Propenol can also react with hydrogen atom to produce vinyl alcohol and methyl radical. 1-propenol is produced by the decomposition of $\text{sC}_4\text{H}_8\text{OH-2}$, which can decompose to produce hydroxyl radical and 2-butene or methyl radical and 1-propenol. Thus, the rate constants for hydrogen atom abstraction from *sec*-butanols and 1-propenol chemistry, and branching ratio of $\text{sC}_4\text{H}_8\text{OH-2}$ decomposition affects the concentration of propionaldehyde. *N*-butyraldehyde, isobutyraldehyde and ethyl methyl ketone are also expected as pyrolysis products of butanol isomers. The highest concentration simulated for these compounds was about 0.2% of *n*-butyraldehyde at 1300 K of *n*-butanol pyrolysis. This result agrees with our experimental results.

4.1.6. Enol chemistry

We included the isomerization of enols to ketones or aldehydes, their reactions with hydrogen atom to produce a methyl radical and a smaller enol, or a hydroxyl radical and an alkene as enol chemistry in the reaction mechanism of butanol isomers. The simulated time-dependent concentration profiles for enols (vinyl alcohol, propen-2-ol and propen-1-ol) are shown in Figs. 2b, 5b, 7b, and 10b. Here, we focus on the case of Fig. 10b. The concentrations of propen-1-ol and vinyl alcohol peak at an early stage of pyrolysis, then begin to decrease, and significant amounts of enols still remain even at 2.5 ms after shock arrival. Propen-1-ol is produced by the β -scission of $\text{iC}_4\text{H}_9\text{OH-1}$, and consumed by the isomerization reaction to propionaldehyde and also reacts with hydrogen atom to produce a vinyl alcohol and a methyl radical. Vinyl alcohol is produced by the reaction of propen-1-ol with hydrogen atom, and consumed by the isomerization reaction to acetaldehyde. At the conditions described in the caption of Fig. 10c, 40% of the acetaldehyde is produced through the isomerization reaction of vinyl alcohol. Thus, the concentration of acetaldehyde in Fig. 10a is influenced by enol chemistry.

4.2. Oxidation

4.2.1. *tert*-Butanol

Figure 12 depicts the effect of equivalence ratio on ignition delay times for *tert*-butanol/oxygen mixtures at average pressures of 1.0 atm reported by Moss et al. [16] The high negative dependence [60,61] of ignition time on oxygen concentration was observed for *tert*-butanol/oxygen mixtures. The model captures quite well the trends and the values of ignition delays. Sensitivity analyses were carried out to investigate reaction sensitivity to ignition time for *tert*-butanol oxidation. Figure 13 depicts sensitivity coefficients as a function of equivalence ratio for mixtures E, F and G at 1450 K and a pressure of 1.0 atm corresponding to conditions in Fig. 12. The water elimination reaction of *tert*-butanol has a higher positive value for stoichiometric than lean conditions, while C–C

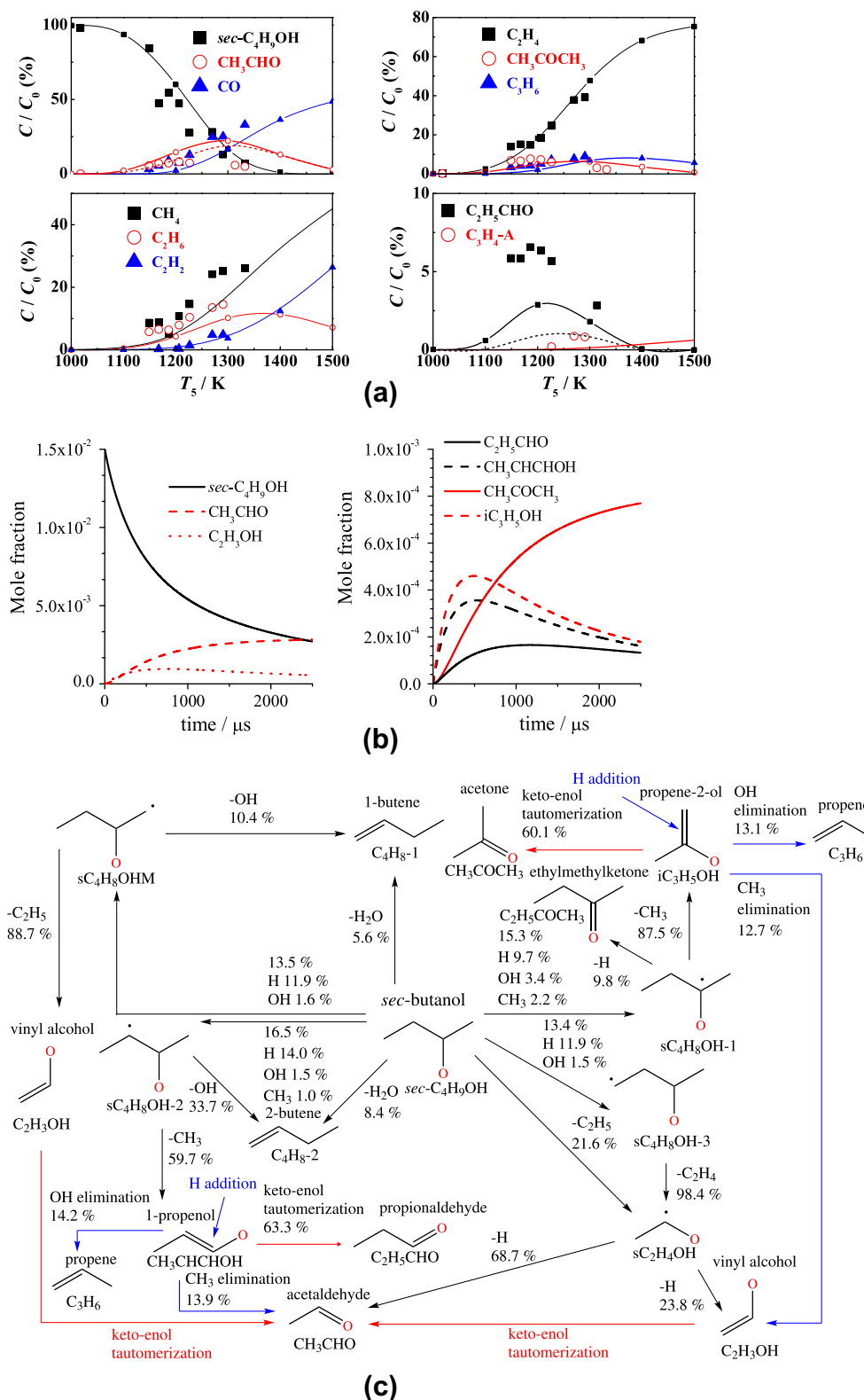


Fig. 7. (a) Normalized species concentration profiles for the pyrolysis of 1.5% *sec*-C₄H₉OH in Ar, effective heating time = 2.6 – 3.0 ms, lines simulation. C₀ and C denote the initial concentrations of *sec*-C₄H₉OH and products, respectively. The dashed line for acetaldehyde, acetone and propionaldehyde stand for their concentration, while solid line represents the summation of them and the corresponding enols. (b) Simulated concentration profiles for 1.5% *sec*-C₄H₉OH diluted in Ar at 1300 K and 1.9 atm. (c) Reaction path analysis for *sec*-C₄H₉OH pyrolysis. Shock conditions: 1.5% *sec*-C₄H₉OH diluted in Ar, 1300 K, 1.9 atm, 20% consumption.

bond scission of *tert*-butanol has a higher negative value for lean than stoichiometric condition. Figure 13 also depicts sensitivity coefficient as a function of temperature for mixture E at 1.0 atm

corresponding to conditions in Fig. 12. C—C bond scission and water elimination reactions of *tert*-butanol have higher negative or positive value at 1450 K than 1800 K.

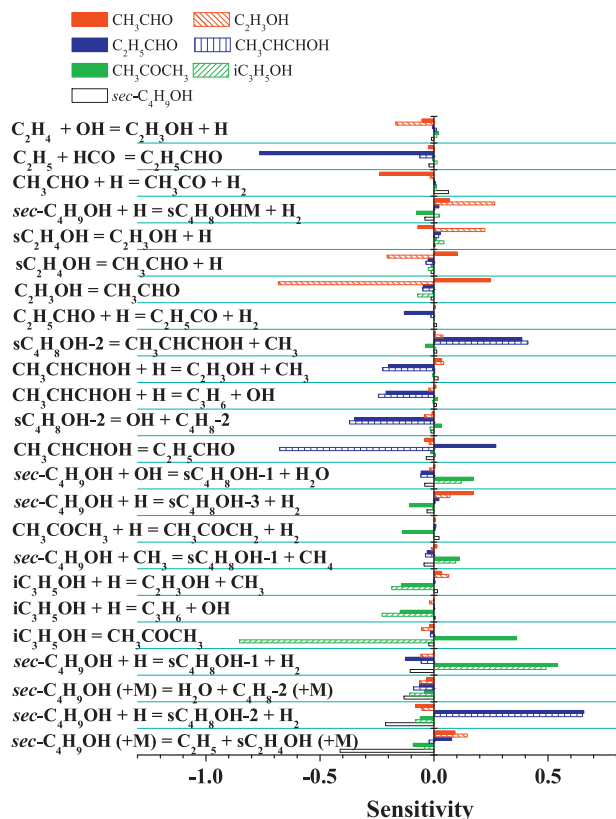


Fig. 8. Sensitivity to species concentration; 1.5% $\text{sec-C}_4\text{H}_9\text{OH}$ diluted in Ar, 1300 K, 1.9 atm, $t_e = 2.6$ ms.

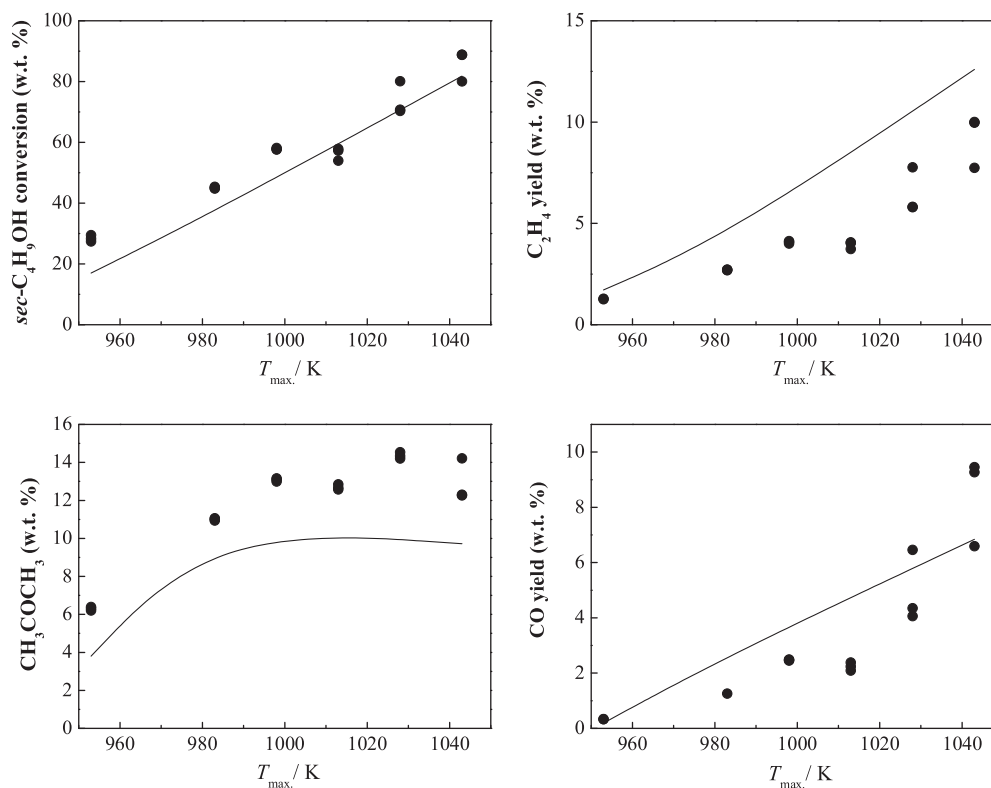


Fig. 9. Conversion and yield profiles for $\text{sec-C}_4\text{H}_9\text{OH}$, ethylene, acetone and carbon monoxide in the pyrolysis reactor at 1.7 atm measured by Van Geem et al. [20]. The simulation line for acetone represents the summation of acetone and propene-2-ol.

4.2.2. *sec*-Butanol

Figure 14 depicts the effect of equivalence ratio and pressure on ignition delay time for 1.0% *sec*-butanol/oxygen mixtures at average pressures of 1.0 and 3.5 atm reported by Moss et al. [16] and measured in *this work*. At 3.5 atm, we observe the high negative dependence [60,61] of ignition time on oxygen concentration which is observed at high temperatures ($1250 \leq T \leq 1850$ K) for almost all hydrocarbon and oxygenated hydrocarbon fuels. It can be observed that as the pressure increases, which is an effective increase in concentration, the ignition delay time becomes faster. Sensitivity analyses were carried out to investigate reaction sensitivity to ignition time for *sec*-butanol oxidation. Figure 15 depicts sensitivity coefficients as a function of equivalence ratio for mixtures H, I and J at 1450 K and a pressure of 3.5 atm corresponding to conditions in Fig. 14.

The radical branching reaction, $\dot{\text{H}} + \text{O}_2 = \ddot{\text{O}} + \dot{\text{OH}}$, has a higher value for rich than stoichiometric and lean conditions, while C—C bond scission of *sec*-butanol to form ethyl and $\text{sC}_2\text{H}_4\text{OH}$ radicals does higher value for lean and stoichiometric than rich conditions. Figure 15 also depicts sensitivity coefficients as a function of pressure for mixture I at 1450 K corresponding to conditions in Fig. 14. Sensitivity coefficients at both pressures do not show large differences. In addition Fig. 15 depicts sensitivity coefficients as a function of temperature for mixture J at 3.5 atm corresponding to conditions in Fig. 14. The radical branching reaction, $\dot{\text{H}} + \text{O}_2 = \ddot{\text{O}} + \dot{\text{OH}}$, has a higher value at 1450 K than 1250 K, while C—C bond scission of *sec*-butanol to form ethyl and $\text{sC}_2\text{H}_4\text{OH}$ radicals, and methyl and hydroperoxy radicals reaction to produce methoxy and hydroxyl radicals have higher value at 1250 K compared to 1450 K.

4.2.3. Isobutanol

Figure 16 depicts the effect of equivalence ratio and pressure on ignition delay time for 1.0% isobutanol/oxygen mixtures at average

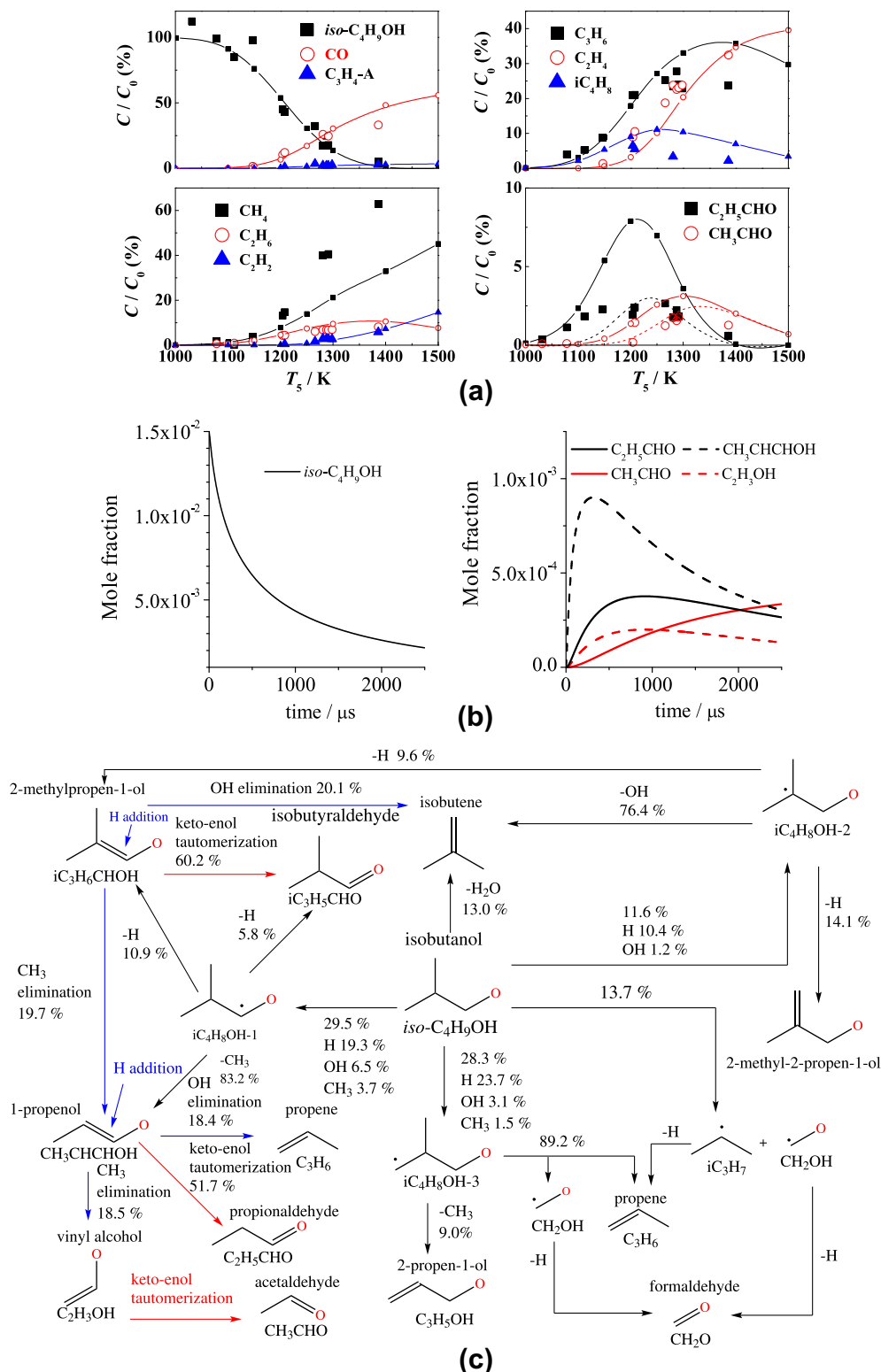


Fig. 10. (a) Species profiles for the pyrolysis of 1.5% $iso-C_4H_9OH$ diluted in Ar, effective heating time = 2.6 – 3.0 ms, lines simulation. C_0 and C denote the initial concentrations of $iso-C_4H_9OH$ and products, respectively. The dashed line for acetaldehyde and propionaldehyde stand for their concentration, while solid line represents the summation of them and the corresponding enols (b) Simulated concentration profiles for 1.5% $iso-C_4H_9OH$ diluted in Ar at 1300 K and 1.9 atm. (c) Reaction path analysis for $iso-C_4H_9OH$ pyrolysis. Shock conditions: 1.5% $iso-C_4H_9OH$ diluted in Ar, 1300 K, 1.9 atm, 20% consumption.

pressures of 1.0 and 3.5 atm reported by Moss et al. [16] and measured in *this work*. At 3.5 atm, we observe the high negative dependence [60,61] of ignition time on oxygen concentration which is

observed at high temperatures ($1250 \leq T \leq 1850$ K) for almost all hydrocarbon and oxygenated hydrocarbon fuels. It can be observed that as the pressure increases, which is an effective increase in

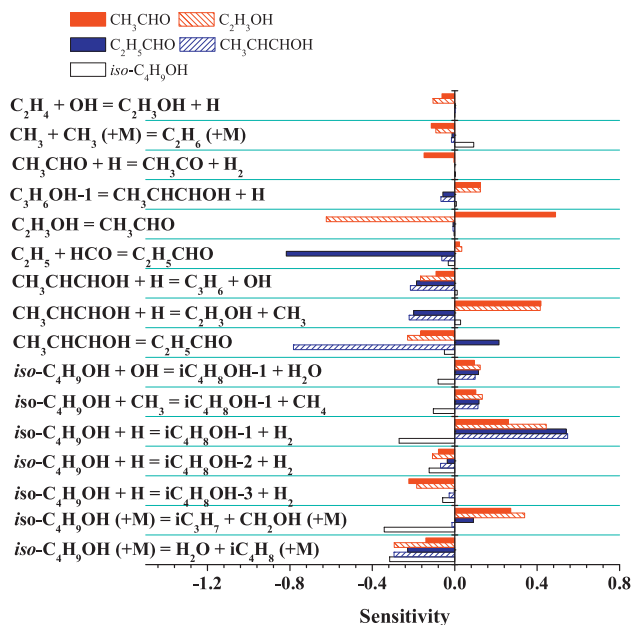


Fig. 11. Sensitivity to species concentration; 1.5% *iso*-C₄H₉OH diluted in Ar, 1300 K, 1.9 atm, $t_e = 2.6$ ms.

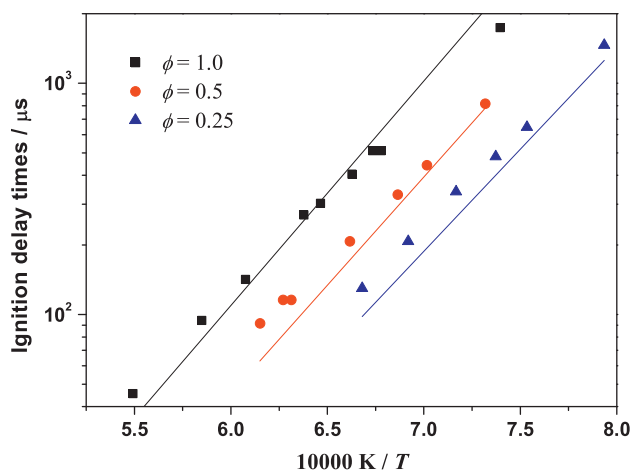


Fig. 12. Ignition delay times at 1.0% *tert*-C₄H₉OH in Ar; ■, $\phi = 1.0$ (6% O₂), ●, $\phi = 0.5$ (12% O₂), ▲, $\phi = 0.25$ (24% O₂) at $p_5 = 1.0$ atm measured by Moss et al. [16]. Lines show the simulation results.

concentration, the ignition delay time becomes faster. Sensitivity analyses were carried out to investigate reaction sensitivity to ignition time for isobutanol oxidation. Figure 17 depicts sensitivity coefficients as a function of equivalence ratio for mixtures K, L and M at 1450 K and a pressure of 3.5 atm corresponding to conditions in Fig. 16.

The radical branching reaction, $\dot{H} + O_2 = \ddot{O} + \dot{OH}$, has a higher value for rich than stoichiometric and lean conditions. C–C bond scission of isobutanol to form an isopropyl and hydroxymethyl radicals has a higher negative value for lean than stoichiometric and rich conditions, while water elimination of isobutanol has a higher positive value for rich than stoichiometric and lean conditions. Figure 17 also depicts sensitivity coefficients as a function of pressure for mixture L at 1450 K corresponding to conditions in Fig. 16. Sensitivity coefficients at both pressures do not show large differences. In addition, Fig. 17 depicts sensitivity coefficients as a function of temperature for mixture M at 3.5 atm corresponding to conditions in Fig. 16. The radical branching reaction,

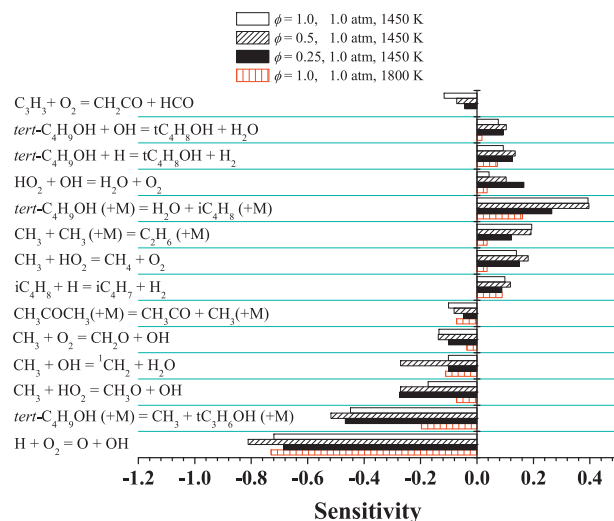


Fig. 13. Sensitivity coefficients showing the effect of equivalence ratio and temperature on ignition time, for 1% *tert*-C₄H₉OH corresponding to the conditions in Fig. 12.

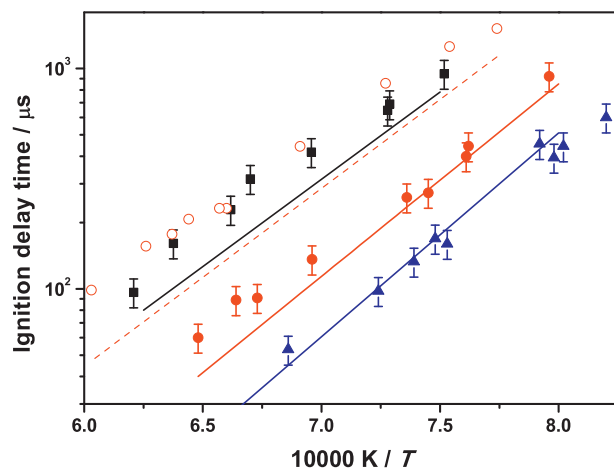


Fig. 14. Ignition delay times at 1.0% *sec*-C₄H₉OH in Ar; ■, $\phi = 2.0$ (3% O₂), ●, $\phi = 1.0$ (6% O₂), ▲, $\phi = 0.5$ (12% O₂) at $p_5 = 3.5$ atm obtained in this work. ○, $\phi = 1.0$ (6% O₂) at $p_5 = 1.0$ atm measured by Moss et al. [16]. Solid and dashed lines show the simulation results at 3.5 and 1.0 atm, respectively.

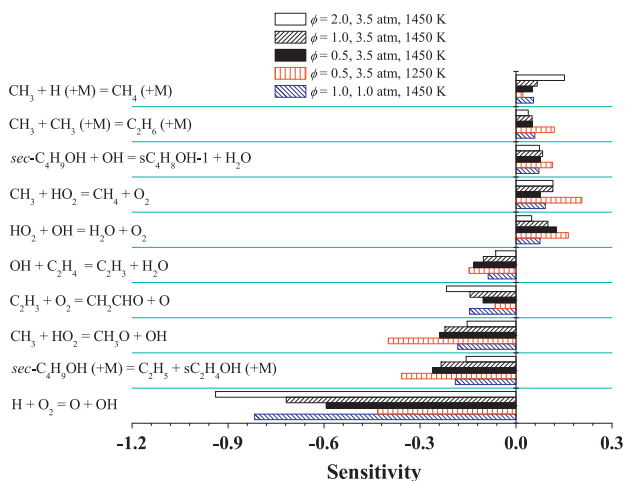


Fig. 15. Sensitivity coefficients showing the effect of equivalence ratio on ignition time for 1% *sec*-C₄H₉OH in Ar corresponding to the conditions in Fig. 14.

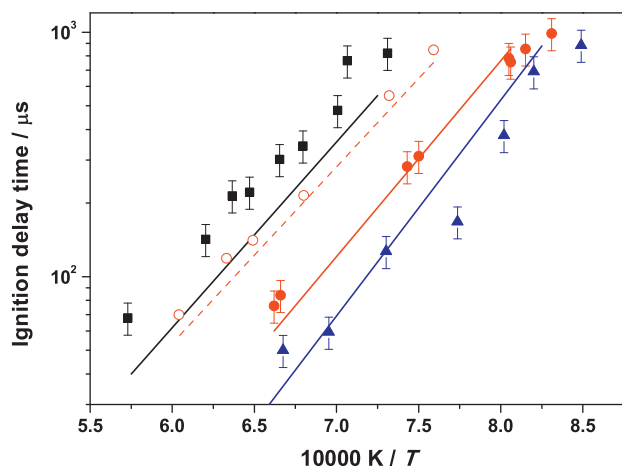


Fig. 16. Ignition delay times at 1.0% *iso*-C₄H₉OH in Ar; ■, $\phi = 2.0$ (3% O₂), ●, $\phi = 1.0$ (6% O₂), ▲, $\phi = 0.5$ (12% O₂) at $p_5 = 3.5$ atm obtained in *this work*. ○, $\phi = 1.0$ (6% O₂) at $p_5 = 1.0$ atm measured by Moss et al. [16]. Solid and dashed lines show the simulation results at 3.5 and 1.0 atm, respectively.

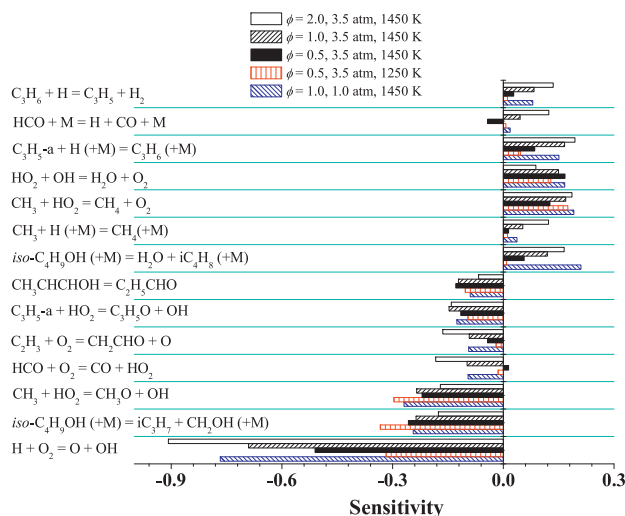


Fig. 17. Sensitivity coefficients showing the effect of equivalence ratio on ignition time for 1% *iso*-C₄H₉OH in Ar corresponding to the conditions in Fig. 16.

$\dot{H} + O_2 = \dot{O} + \dot{OH}$, has a higher value at 1450 K than 1250 K, while C—C bond scission of isobutanol to form an isopropyl and hydroxymethyl radicals, and methyl and hydroperoxy radicals reaction to produce methoxy and hydroxyl radicals have higher value at 1250 K than 1450 K.

5. Conclusions

A comprehensive set of experiments in two different shock tubes has generated a wealth of data describing the thermal decomposition and oxidation of butanol isomers. Thus, species profiles for acetaldehyde, 1-butene, propene, allene (C₃H₄-a), ethane, ethylene, acetylene, methane as well as carbon monoxide were obtained for 1.5% *n*-butanol in the SPST by GC. Species profiles for acetone, isobutene, propene, allene (C₃H₄-a), propyne (C₃H₄-p), ethane, ethylene, acetylene, methane as well as carbon monoxide were obtained for 1.0% *tert*-butanol. Species profiles for acetaldehyde, acetone, propionaldehyde, propene, allene (C₃H₄-a), ethane, ethylene, acetylene, methane as well as carbon monoxide were obtained for 1.5% *sec*-butanol. Species profiles for

acetaldehyde, propionaldehyde, isobutene, propene, allene (C₃H₄-a), ethane, ethylene, acetylene, methane as well as carbon monoxide were obtained for 1.5% isobutanol. The little-known chemistry of enols is included in *this work* to explain the production of aldehydes and acetone.

Ignition delays for 1.0% *sec*-butanol or isobutanol, oxygen and argon mixtures, ranging from fuel lean to rich ($\phi = 0.5, 1.0$ and 2.0), at 3.5 atm were newly obtained. The reaction mechanism constructed in *this work* successfully explains not only our ignition delay data and species concentration profiles but also ignition delay data reported earlier by Moss et al. [16] for *tert*-butanol, *sec*-butanol and isobutanol at 1 atm and species concentration profiles reported by Harper et al. [21] and Van Geem et al. [20] for *n*-butanol and *sec*-butanol at 1.7 atm.

Sensitivity analyses showed that unimolecular water elimination and C—C bond scission reactions of butanol and hydrogen abstraction reactions from butanol by hydrogen atom, methyl and hydroxyl radicals affect the ignition delay times in oxidation as well as the concentration of reactants and products in pyrolysis. On the other hand, C—O and C—H bond scission reactions of butanol do not affect the experimental results under our experimental conditions.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2012.02.008.

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